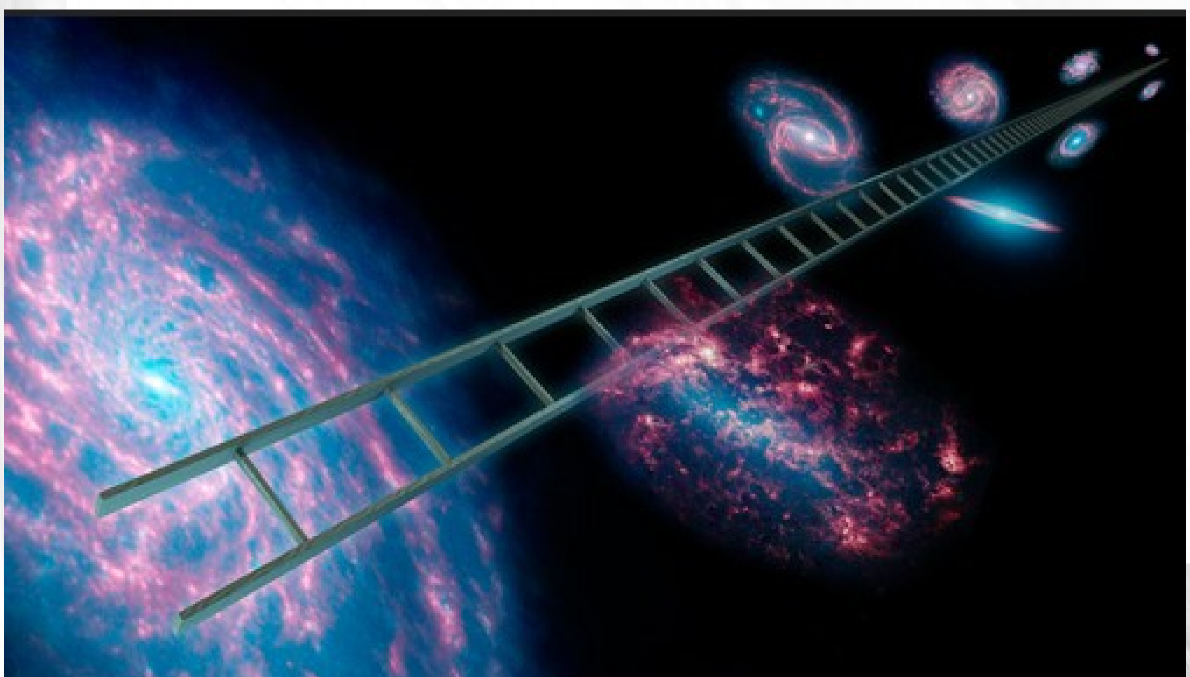
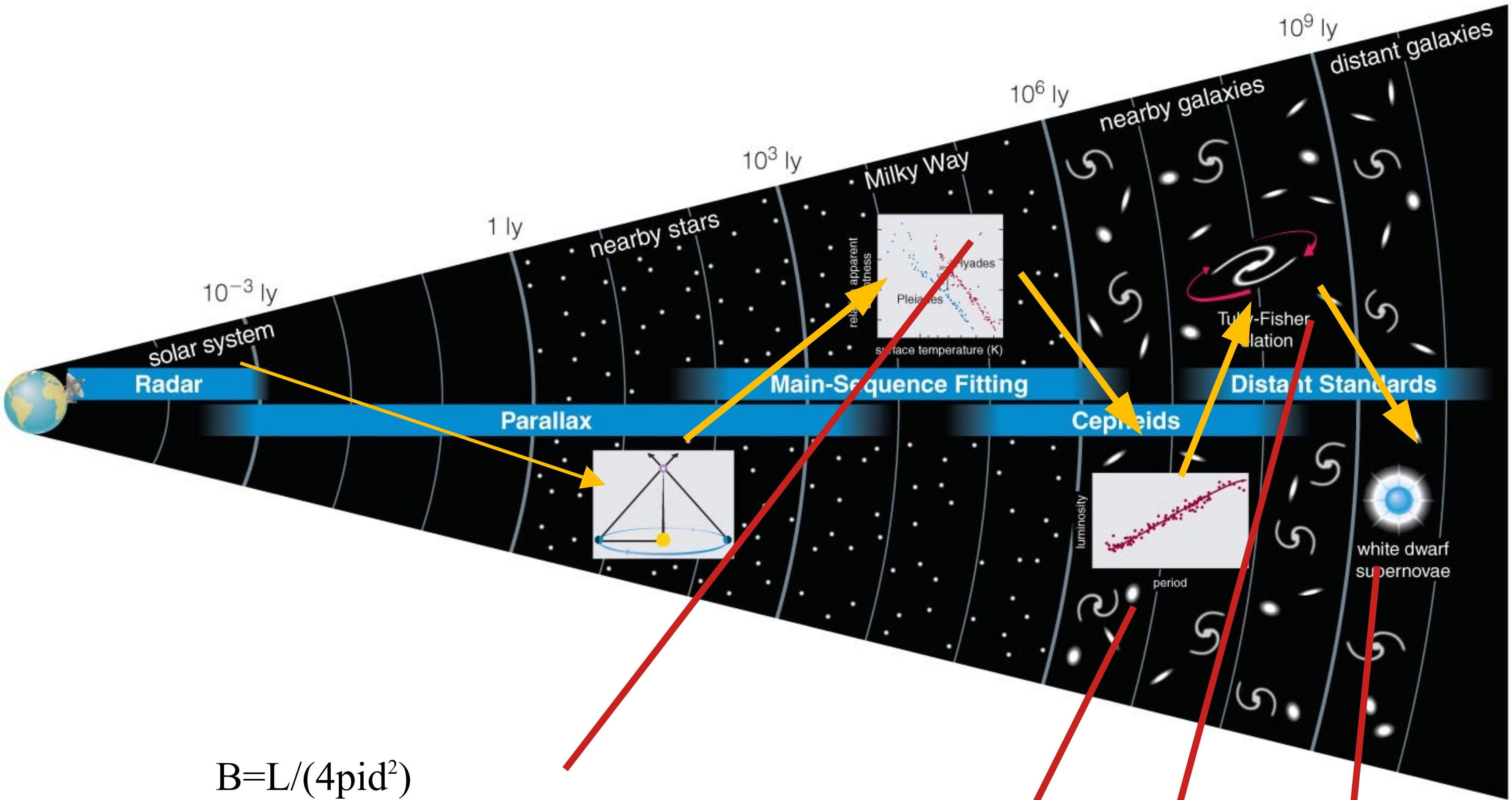
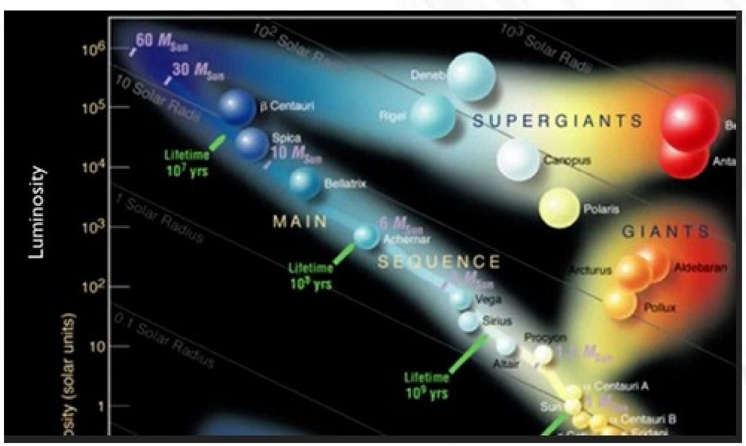
How do we measure the distances to stars, galaxies..?
How to we measure the age of objects in the Universe?
The distance ladder

Astronomers use various methods to measure relative distances in the Universe, depending upon the object being observed. Collectively these techniques are known as the cosmic distance ladder. It's called a ladder for good reason — each rung or measurement technique relies upon the previous step for calibration.




10^{-3} ly
1 ly
10^{3} ly
10^{6} ly
10^{9} ly
solar system
nearby stars
Milky Way
nearby galaxies
distant galaxies
Radar
Parallax
Main-Sequence Fitting
Cepheids
Distant Standards
Tully-Fisher relation
white dwarf supernovae
luminosity
period
surface temperature (K)
Pleiades
Hyades

B=L/(4pid^2)

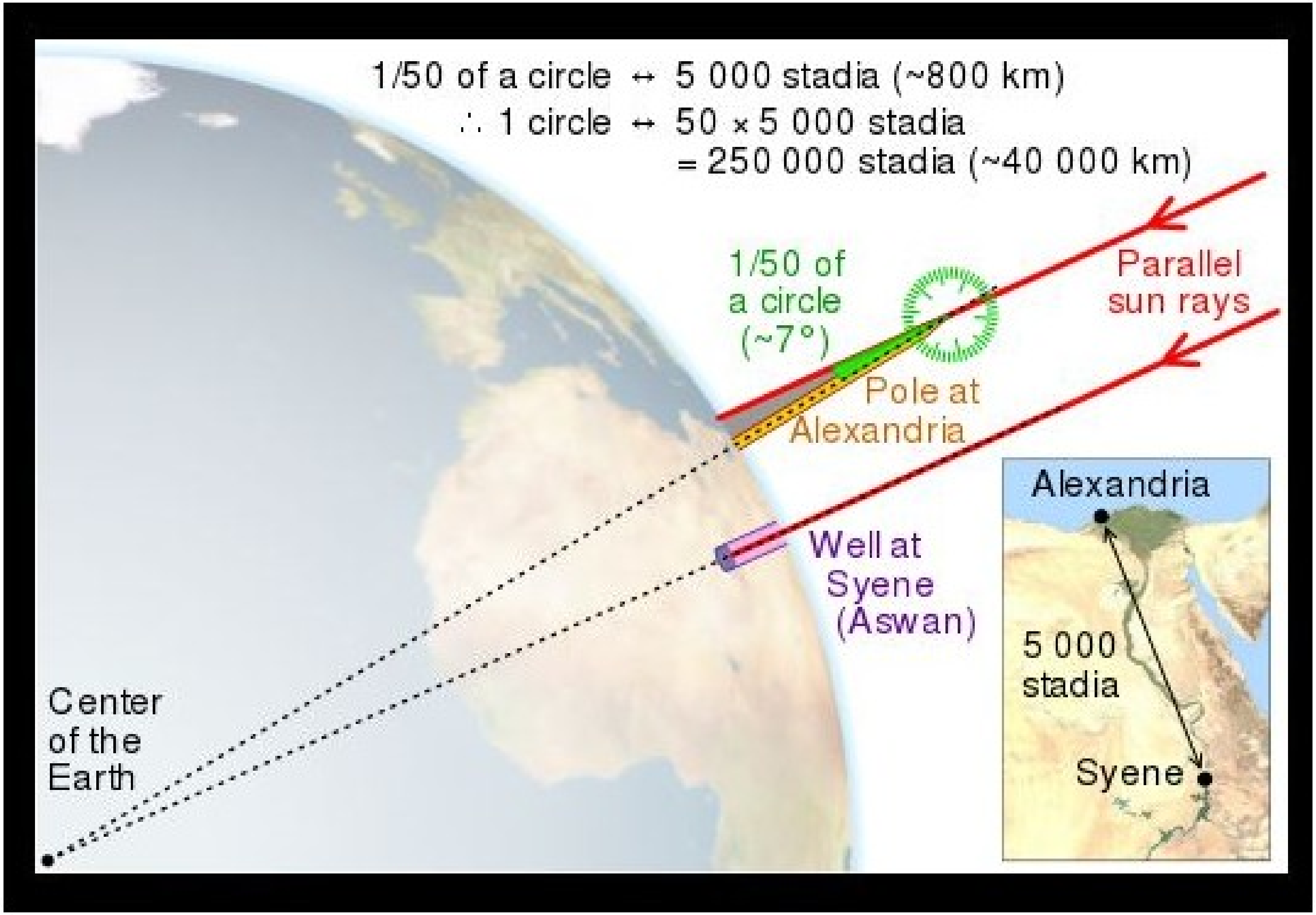

Luminosity
SUPERGIANTS
GIANTS
MAIN
SEQUENCE
60 M_{Sun}
30 M_{Sun}
10 M_{Sun}
6 M_{Sun}
Deneb
Rigel
Canopus
Polaris
β Centauri
Spica
Bellatrix
Achernar
Arcturus
Aldebaran
Pollux
Vega
Sirius
Procyon
Altair
Sun
Lifetime 10^7 yrs
Lifetime 10^8 yrs
Lifetime 10^9 yrs
10 Solar Radii
10^2 Solar Radii
10^3 Solar Radii
1 Solar Radius
0.1 Solar Radius


If we know how luminous the candle is and how bright it appears we can calculate how far away it is

Movie car Sagan

Measuring the size of the Earth

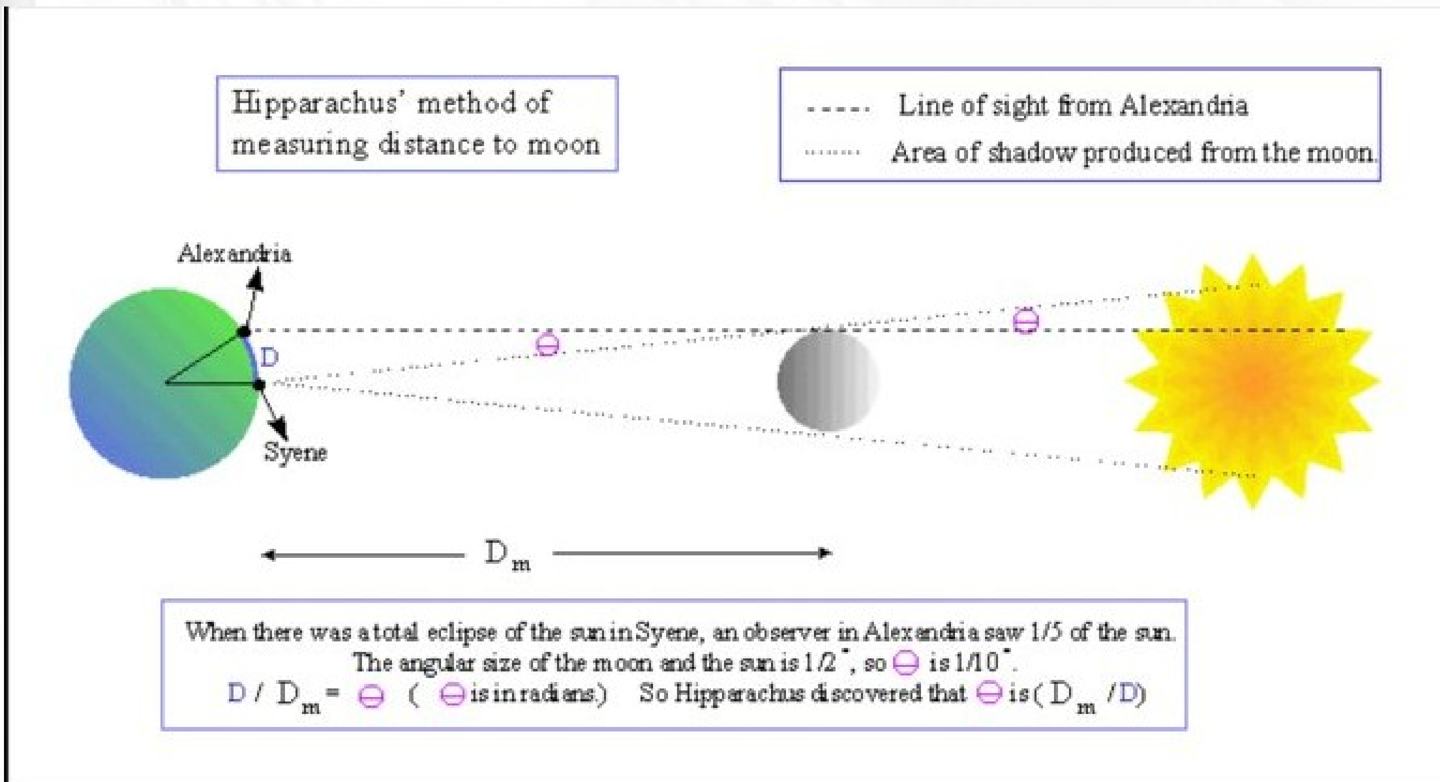
- Eratosthenes of Cyrene - 276 B.C



Eratosthenes inferred that the Earth's radius was 6366 km. Both of these values are very close to the accepted modern values for the Earth's circumference and radius, 40,070 km and 6378 km respectively, which have since been measured by orbiting spacecraft.

Hipparchus distance to the Moon (7% error – 2nd century BCE)

in 190 BC, Hipparchus measured the distance from the Earth to the Moon during a solar eclipse that was a total eclipse at Syene and a partial eclipse at Alexandria. He found 396,103 km vs 384,400.

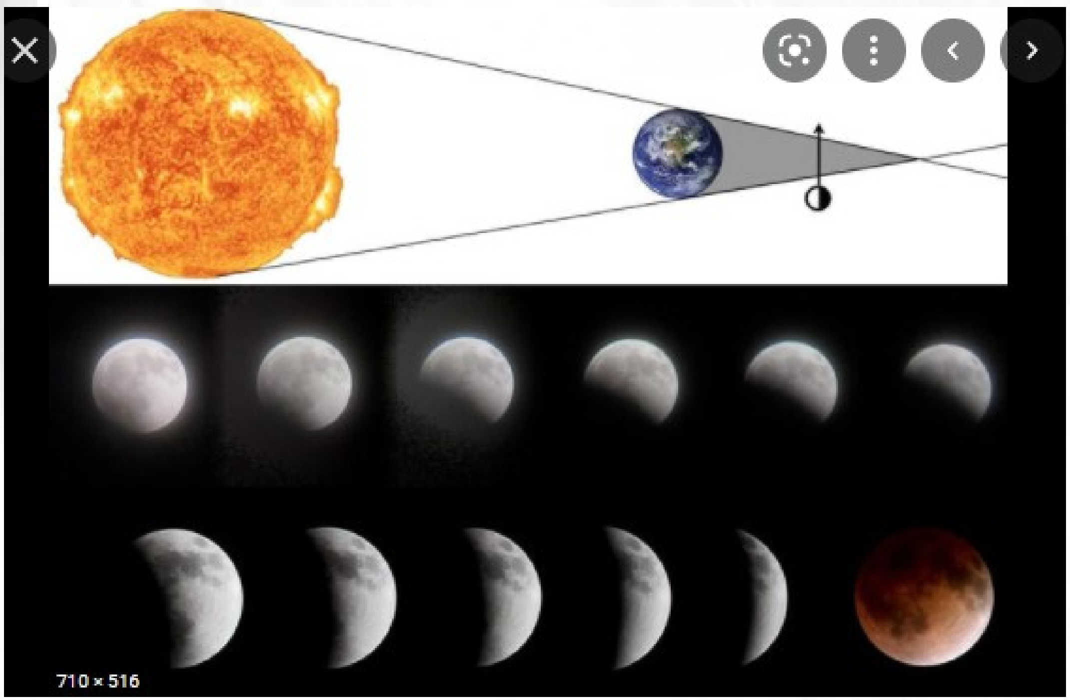


400

Hipparchus's calculations led him to a value for the distance to the moon of **between 59 and 67 earth radii** which is quite remarkable (the correct distance is 60 earth radii). The main reason for his range of values was that he was unable to determine the parallax of the sun, only managing to give an upper value.

See video eclipse by NASA

To find the ratio size Moon vs size Earth (needed to find the distance Earth-Moon)
He used a lunar eclipse. When the Earth casts a shadow on the Moon.

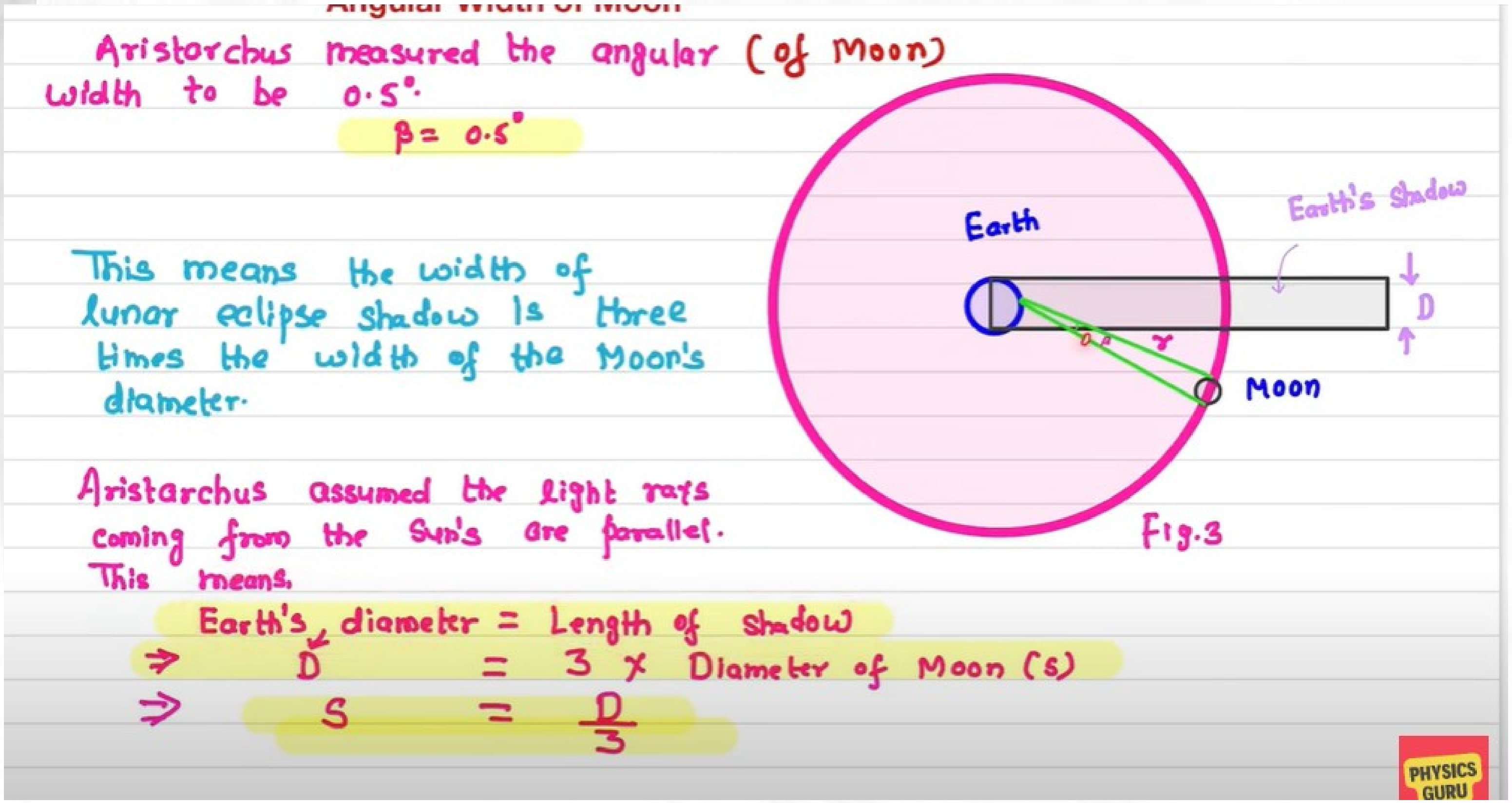


See image eclipse.png in folder.

You can compare the size of the Moon
To the size of the Earth.

Before: Aristarchus

Aristarchus of Samos (310-230 BC). His measurement begins with the observation of a total lunar eclipse. He was off by about 20% for the size of the Moon (3 times less than Earth instead of 3.7).
He was off by 25% about for the distance.
But very smart protocol.

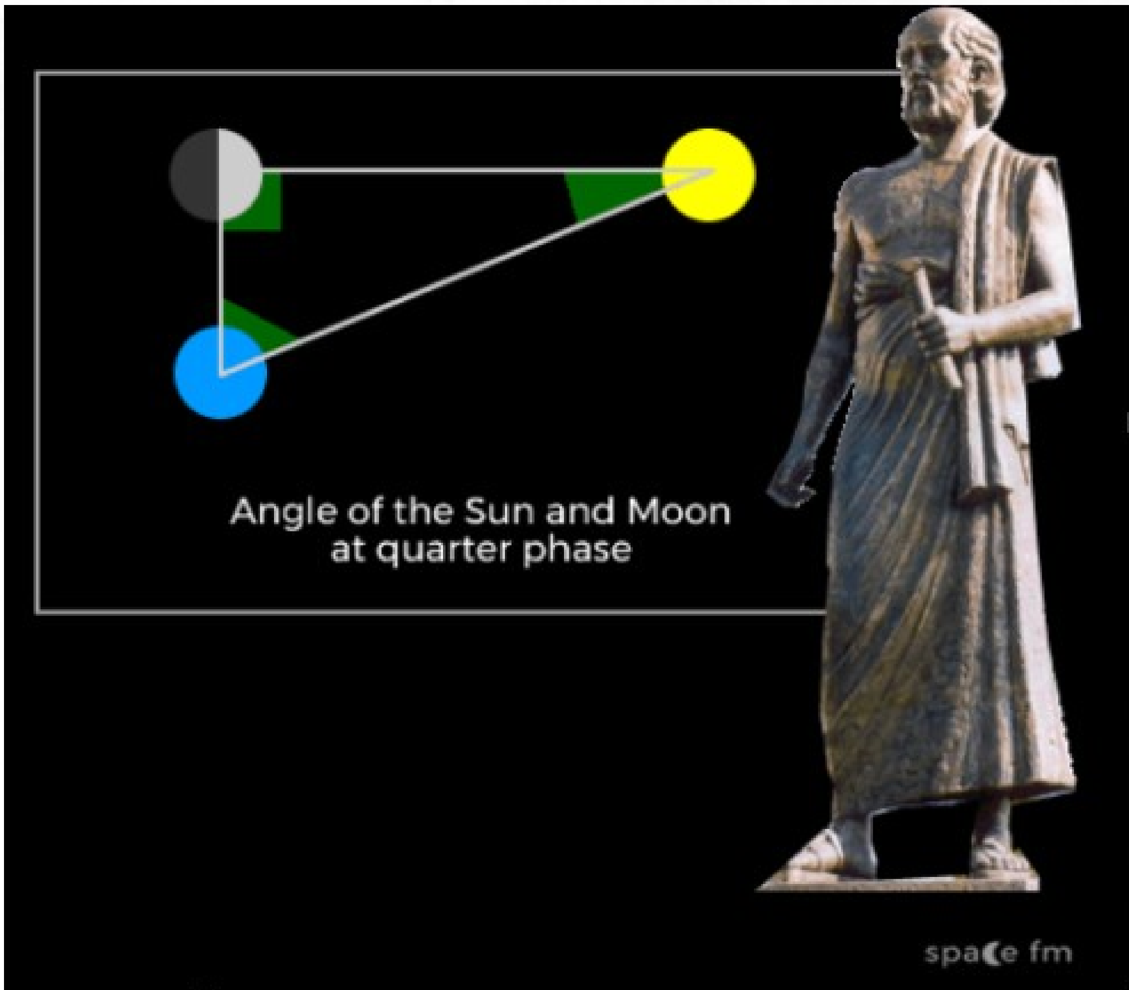


https://www.youtube.com/watch?v=Awu9GSNvcao

ARISTARCHUS

Size and Distances of the Moon and Sun

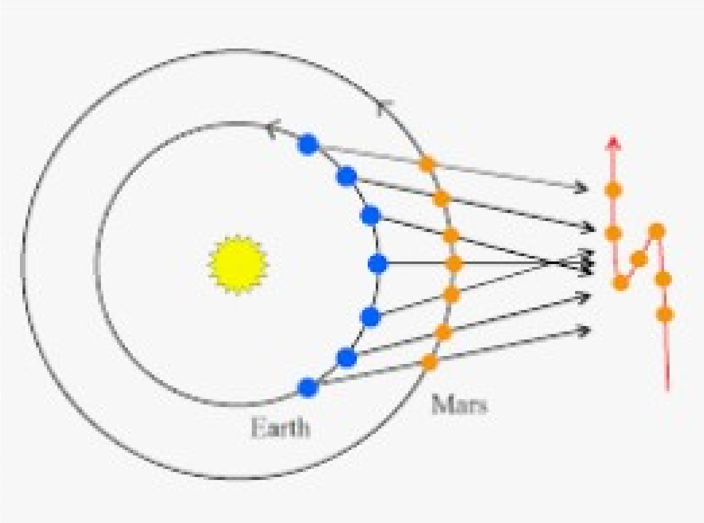
Aristarchus measured the distance to the Sun (knowing the distance to the Moon)
But he was off by a lot.



Result was off by a factor of 1000.
The angle estimated was 87
Instead of 89.83.

Copernicus – heliocentric system relative distance of planets

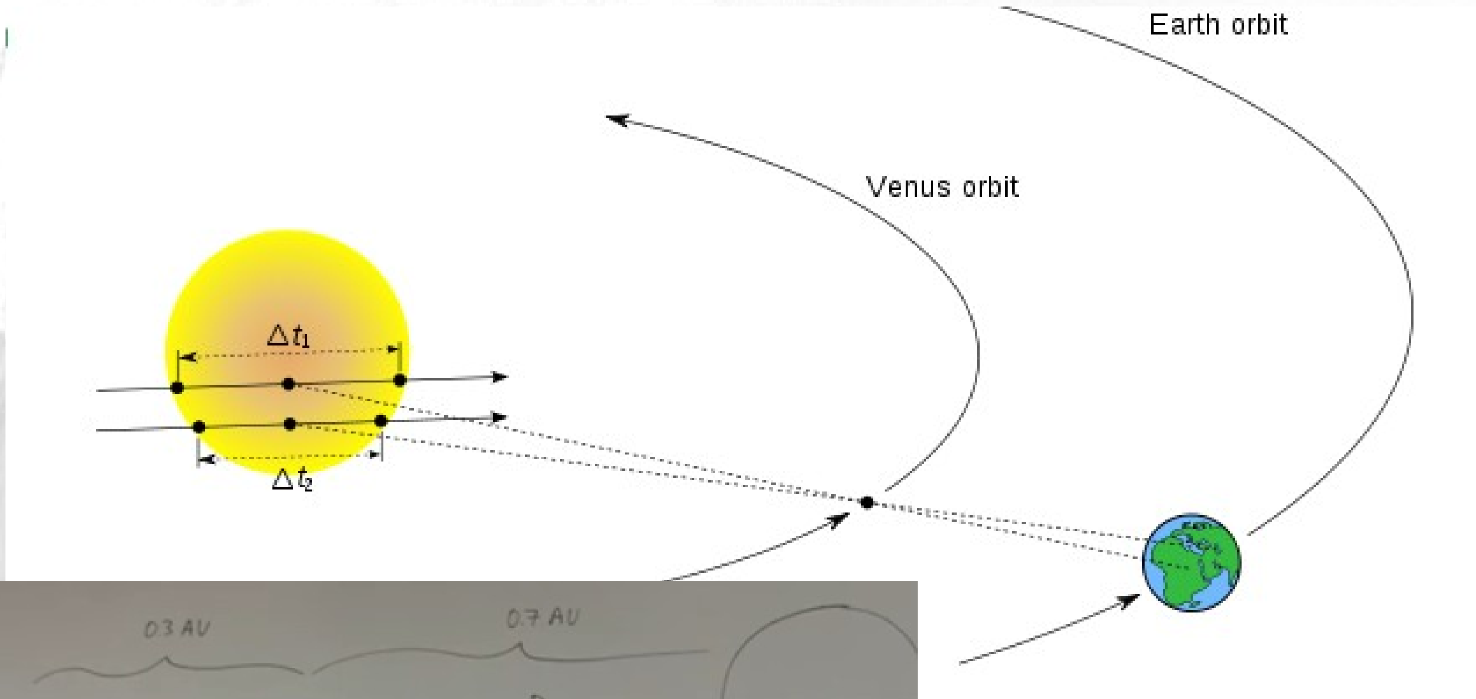
Planet	Copernicus	Modern Value
Mercury	0.38 AU	0.39 AU
Venus	0.72 AU	0.72 AU
Earth	1.00 AU	1.00 AU
Mars	1.52 AU	1.52 AU
Jupiter	5.2 AU	5.20 AU
Saturn	9.2 AU	9.54 AU

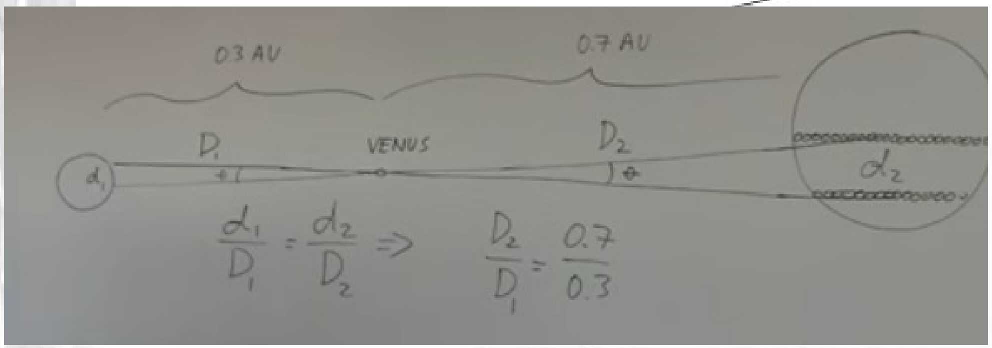


http://www.astronomy.ohio-state.edu/~pogge/Ast161/Unit3/copernicus.html

Jupiter

Absolute distance to Sun





The main aim of Captain James Cook's Endeavour voyage was to observe the transit of Venus from Tahiti in June, 1769. The transit observations were used to calculate the size of the solar system which assisted in nautical navigation.

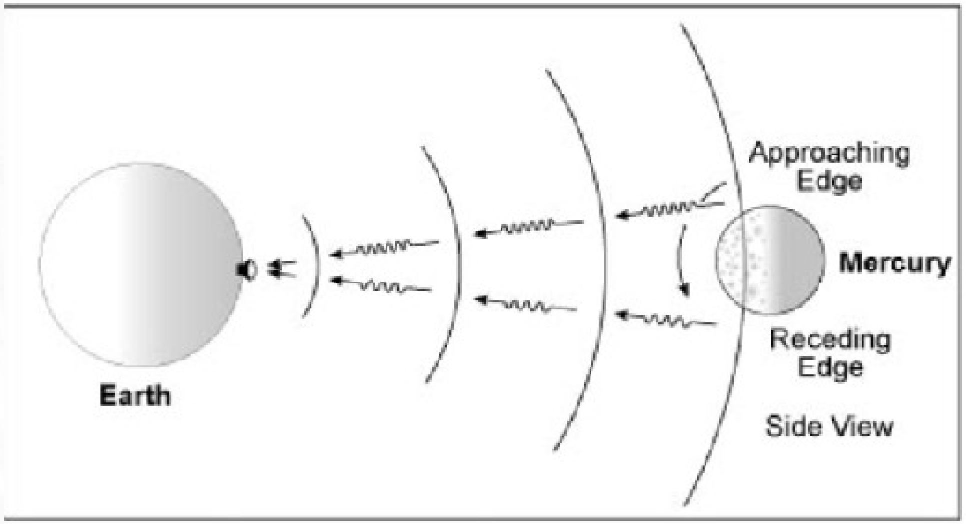
"When Cook got back to Europe mathematicians using Cook's data, calculated the Earth-sun distance to be 153 million kilometers. That was done by German astronomer Johann Georg von Soldner . Another astronomer French astronomer Jerome Lalande tried also but didn't get The best answer.

https://www.youtube.com/watch?v=lET7Qf3WtP4

149,600,000 kilometers (km)

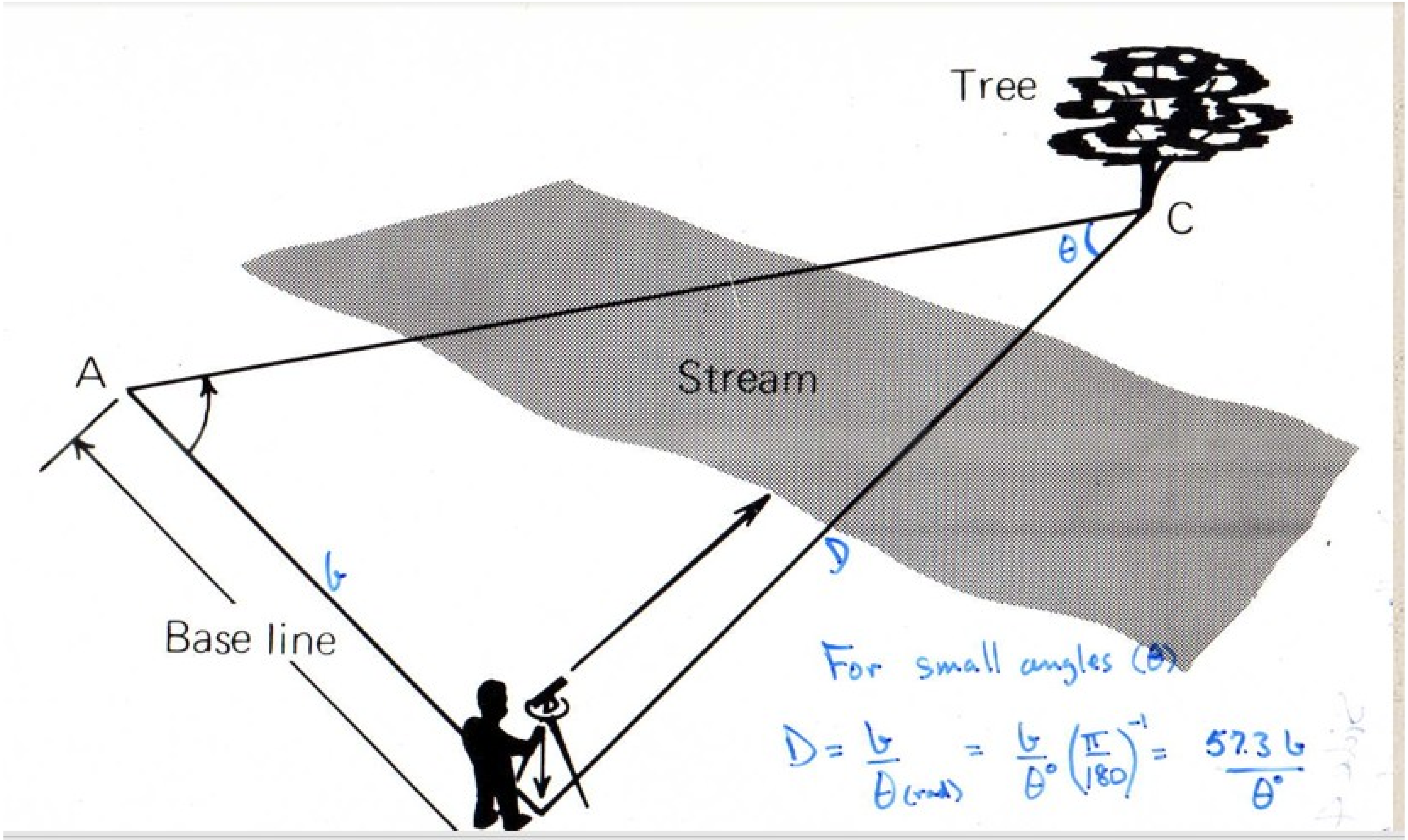
By measuring the time taken for the radar echo to come back, the distance can be calculated, since radio waves travel at the speed of light. Once this Earth-Venus distance is known, the distance between Earth and the Sun can be calculated

https://history.nasa.gov/SP-4218/ch2.htm

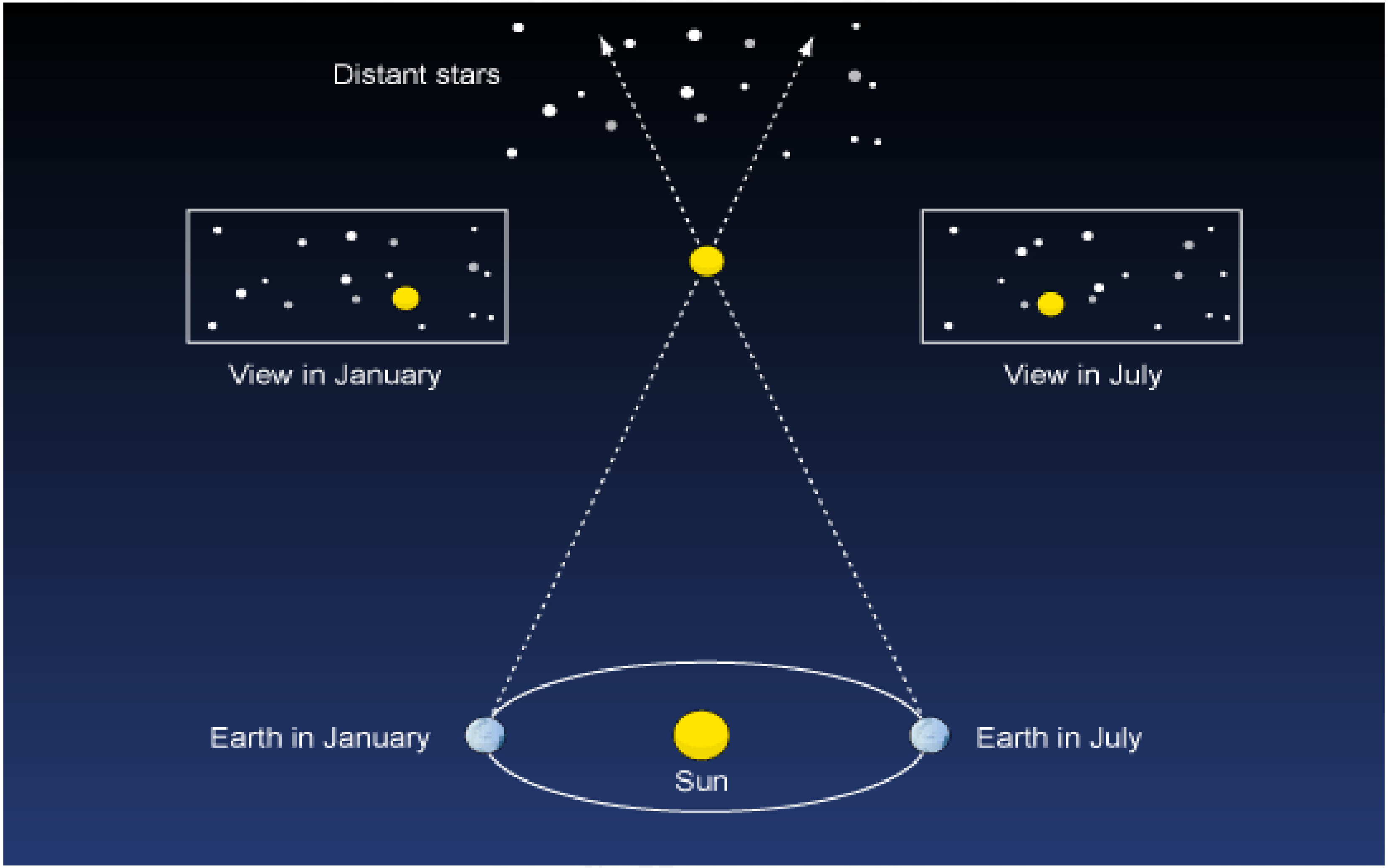


[27] In 1958, MIT's Lincoln Laboratory announced that it had bounced radar waves off Venus. That apparent success was followed by another, but in England, during Venus' next inferior conjunction. In September 1959, investigators at Jodrell Bank announced that they had validated the 1958 results, yet Lincoln Laboratory failed to duplicate them. All uncertainty was swept aside, when the Jet Propulsion Laboratory (JPL) obtained the first unambiguous detection of echoes from Venus in 1961

Measuring distance with triangulation



If we know 1AU → distance to nearby stars using parallax

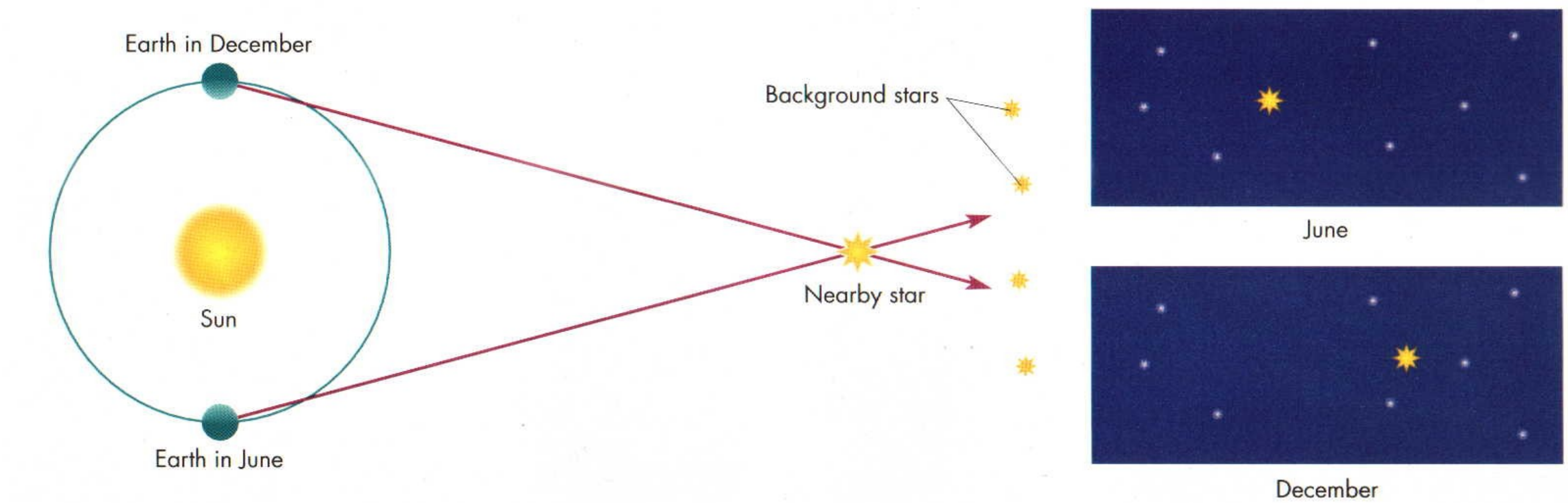


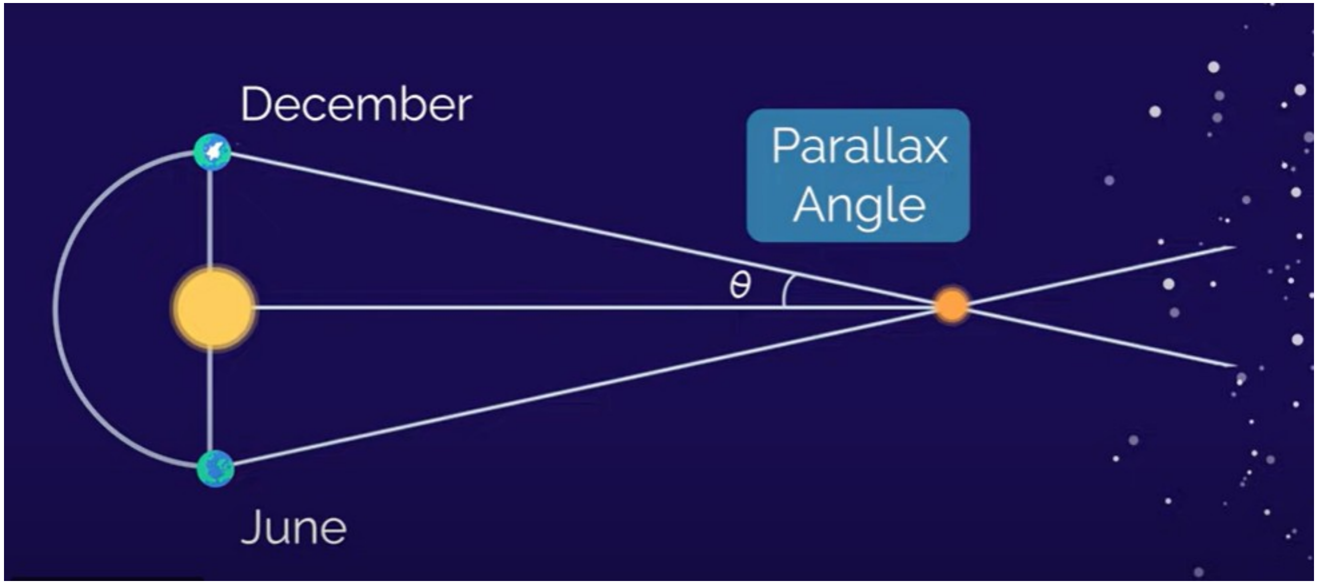
Movies parallax

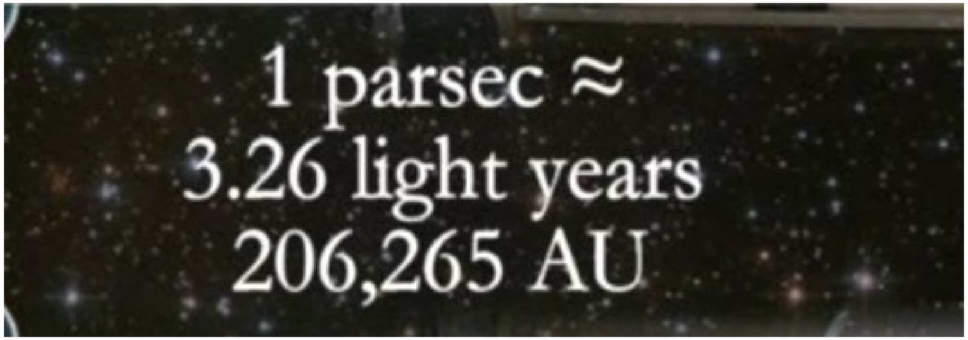
Parallax – direct distances to the stars

d(parsec) = 1/p(arcsec) ; 1 parsec = 3.26 light years

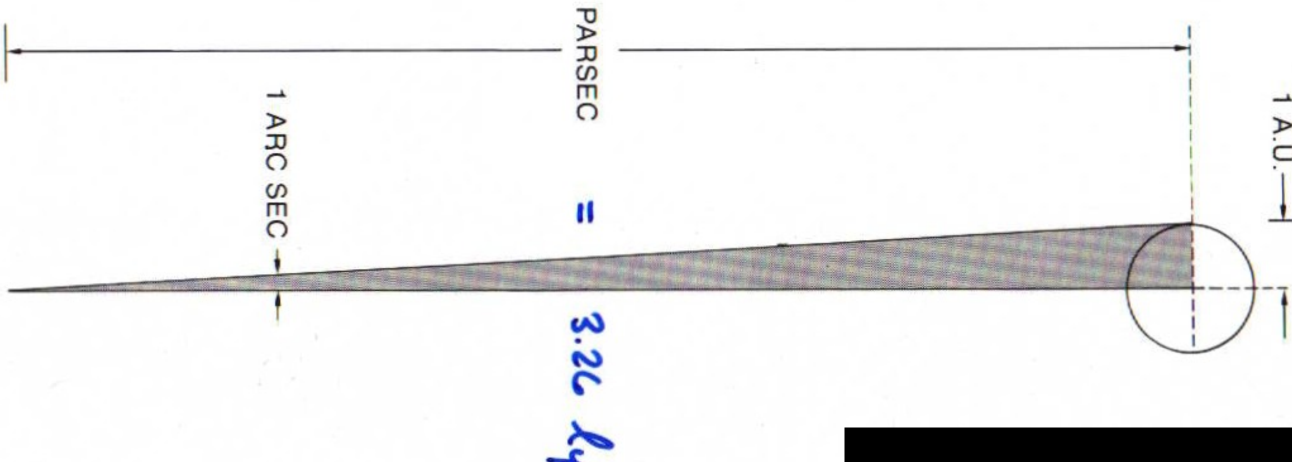
1 arcsecond → 1 dime 2.5 miles away







d (parsecs) $= \frac{1}{p}$ (arc sec)$^{-1}$



$$d\,(\text{pc}) = \frac{1}{p\,(\text{arc sec})}$$

$p = 0.5'' = 1/2''$, $d = 2$ pc

$p = 0.1'' = 1/10''$, $d = 10$ pc

$p = 0.01'' = 1/100''$, $d = 100$ pc

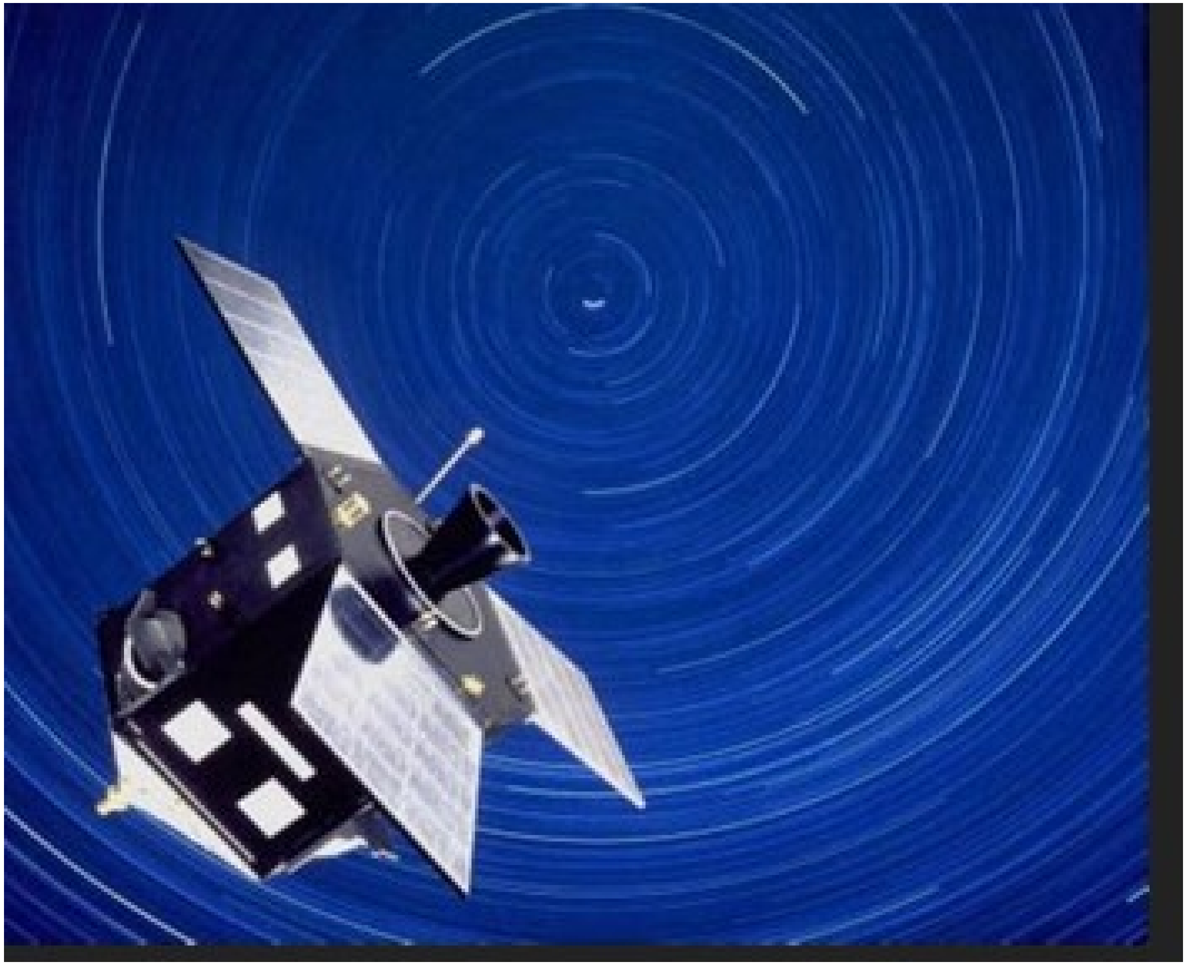
1 parsec ≈ 3.26 light years

Alpha centauri p=0.77 arcseconds
D=1.3 parsecs = 1.33 x 3.26 = 4.2 l.y. away

We can't measure distances larger than 100 l.y. away (and with more errors)
0.01 arcsecs → 100 l.y. Its really good for 10 parsecs. (about 30 l.y.)

Hipparcos was a scientific satellite of the European Space Agency (ESA), launched in 1989 and operated until 1993. It was the first space experiment devoted to precision astrometry

→ catalog for stars 100 l.y away. For larger distances, we use other methods. We use standard candles.

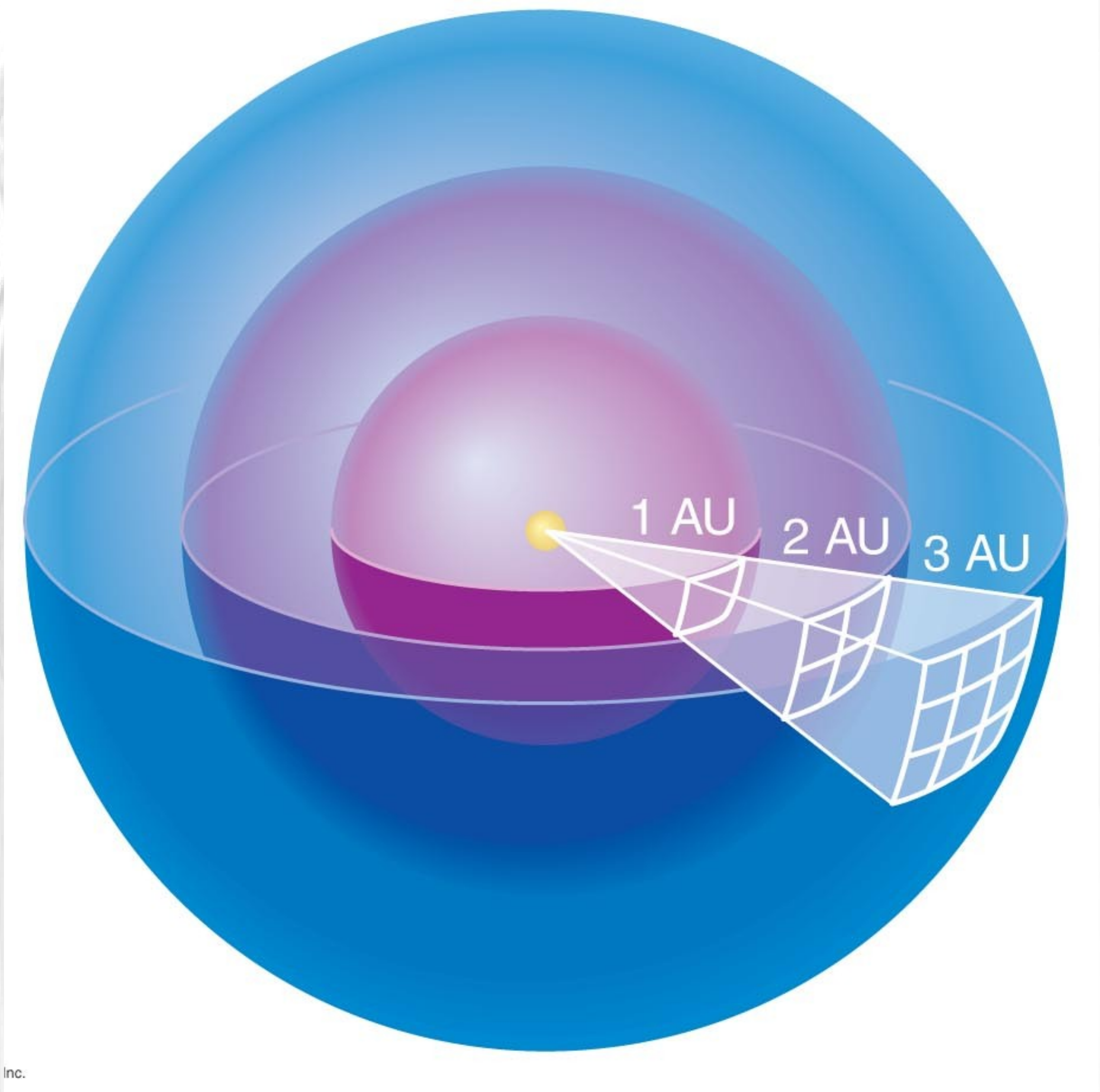


NEXT STEP: CALIBRATE : standard candles.

See example below. If we know how bright a candle is supposed to be and if
We record the brightness (how bright it appears) then we can figure out
The distance. These objects are called standard candles.

Same idea for headlights of cars seen from a distance.
We can estimate the Distance.

The relationship between luminosity and brightness is simple.



Luminosity passing through each sphere is the same.

Area of sphere:

$$4\pi\ (\text{radius})^2$$

Divide luminosity by area to get brightness.

The relationship between apparent brightness and luminosity depends on distance:

$$\text{Brightness} = \frac{\text{Luminosity}}{4\pi\,(\text{distance})^2}$$

We can determine a star's distance if we know its luminosity and can measure its apparent brightness:

$$\text{Distance} = \sqrt{\frac{\text{Luminosity}}{4\pi \times \text{Brightness}}}$$

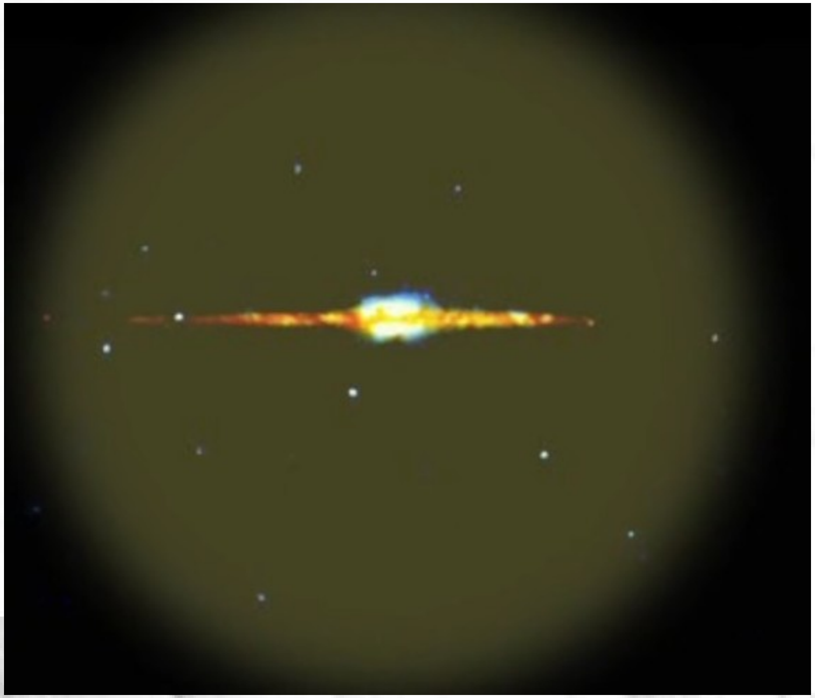
A ***standard candle*** is an object whose luminosity we can determine without measuring its distance.

Next step:using star clusters
Group of stars gravitationally bound.
All the stars have the same distance and age.

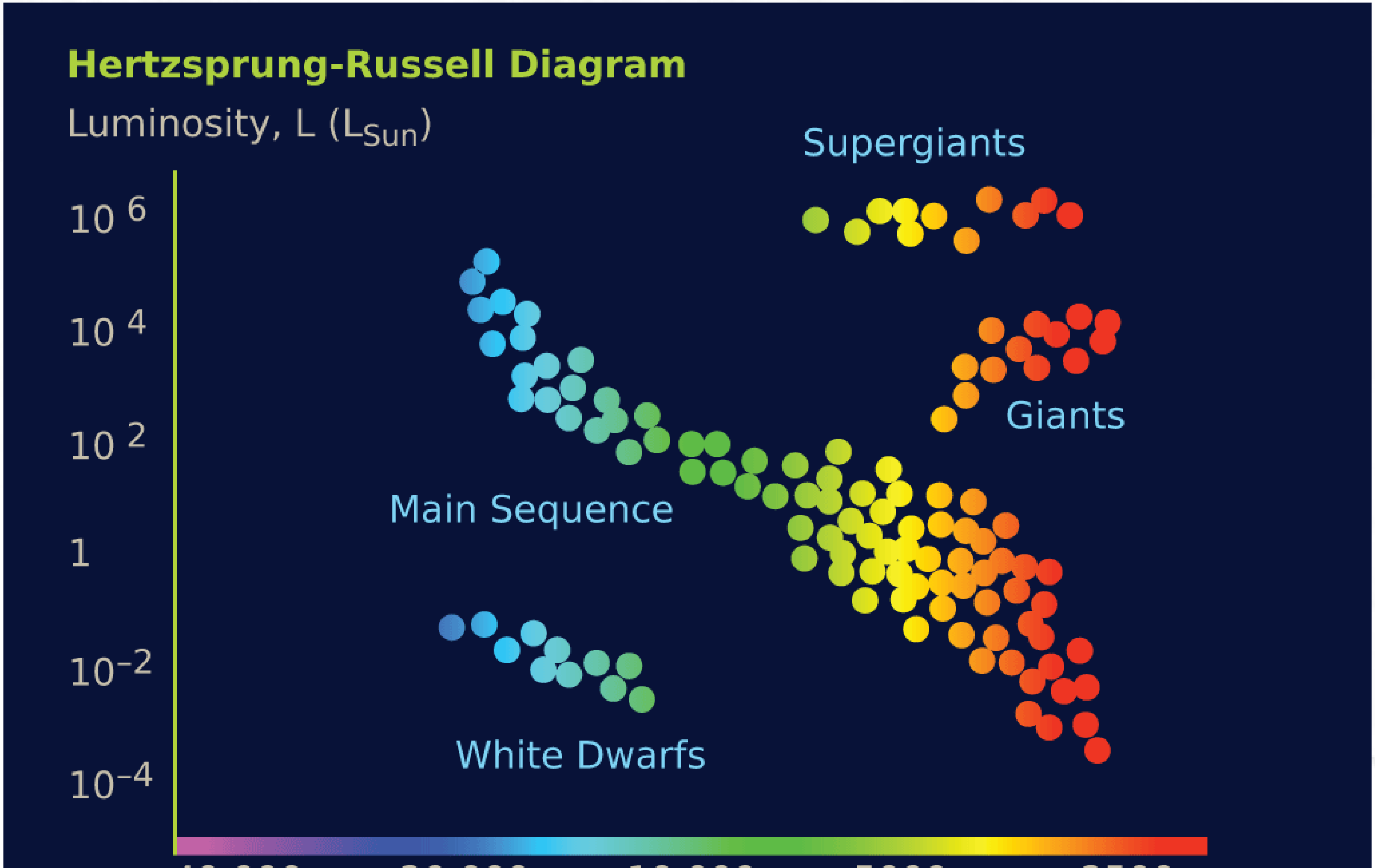
Open clusters

Open clusters found in the arms of spiral galaxies
1000s in ours. Need lots of gas

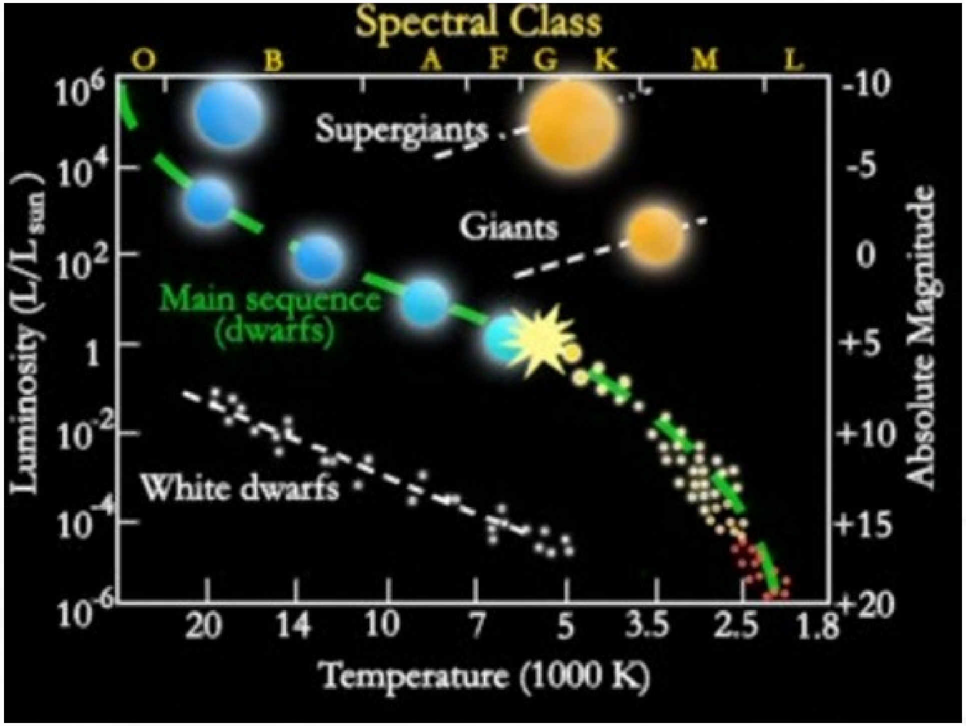
Globular clusters found in elliptical galaxies.
In the halo of galaxies.
20,000 of clusters. Like M87.

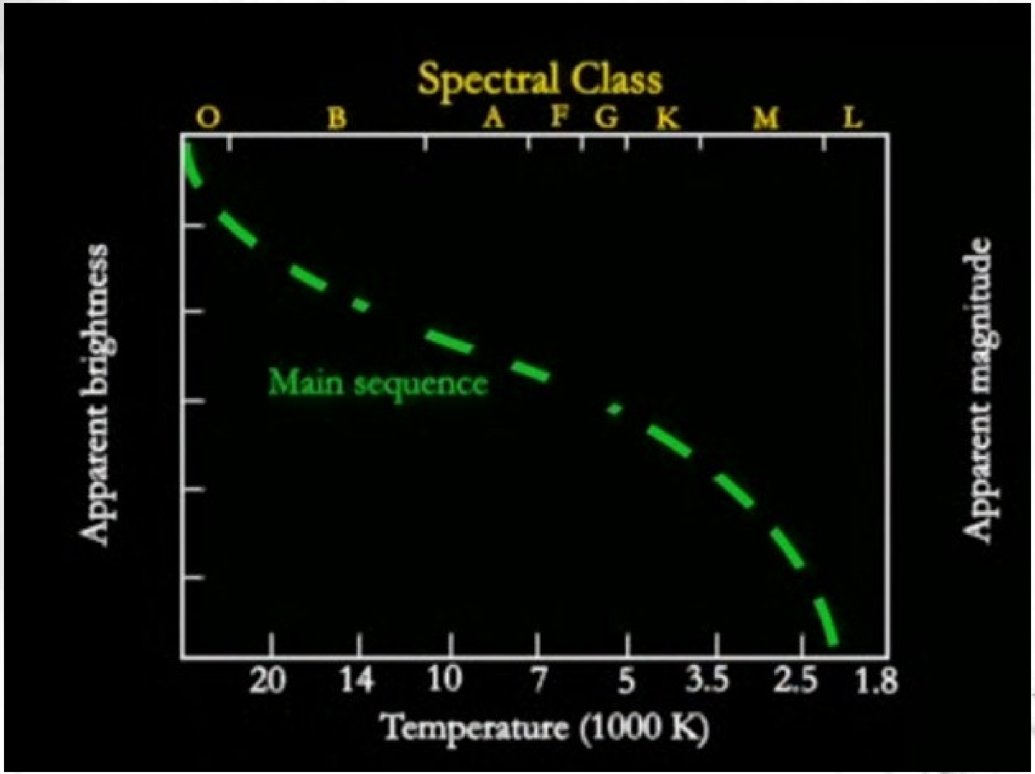


We can plot all the stars from a cluster on a HR diagram.
All the stars are at the same distance and they were all born at the same time
All the HR diagrams (from different clusters, no matter the cluster)
will have the same format
→ main sequence + giants + supergiants+white dwarfs.
If its a old cluster, we will be missing some blue stars and we will have more supergiants.
Ifs its a new cluster, we will have all the stars from main sequence and not many
Giants or white dwarf.



For a given cluster we can plot the stars on a diagram apparent brightness versus temperature. Temperature is induced from the spectra of starts. We look at the main sequence. Luminosity is proportional to Brightness. All the stars are shifted by the same amount. This shift → distance.

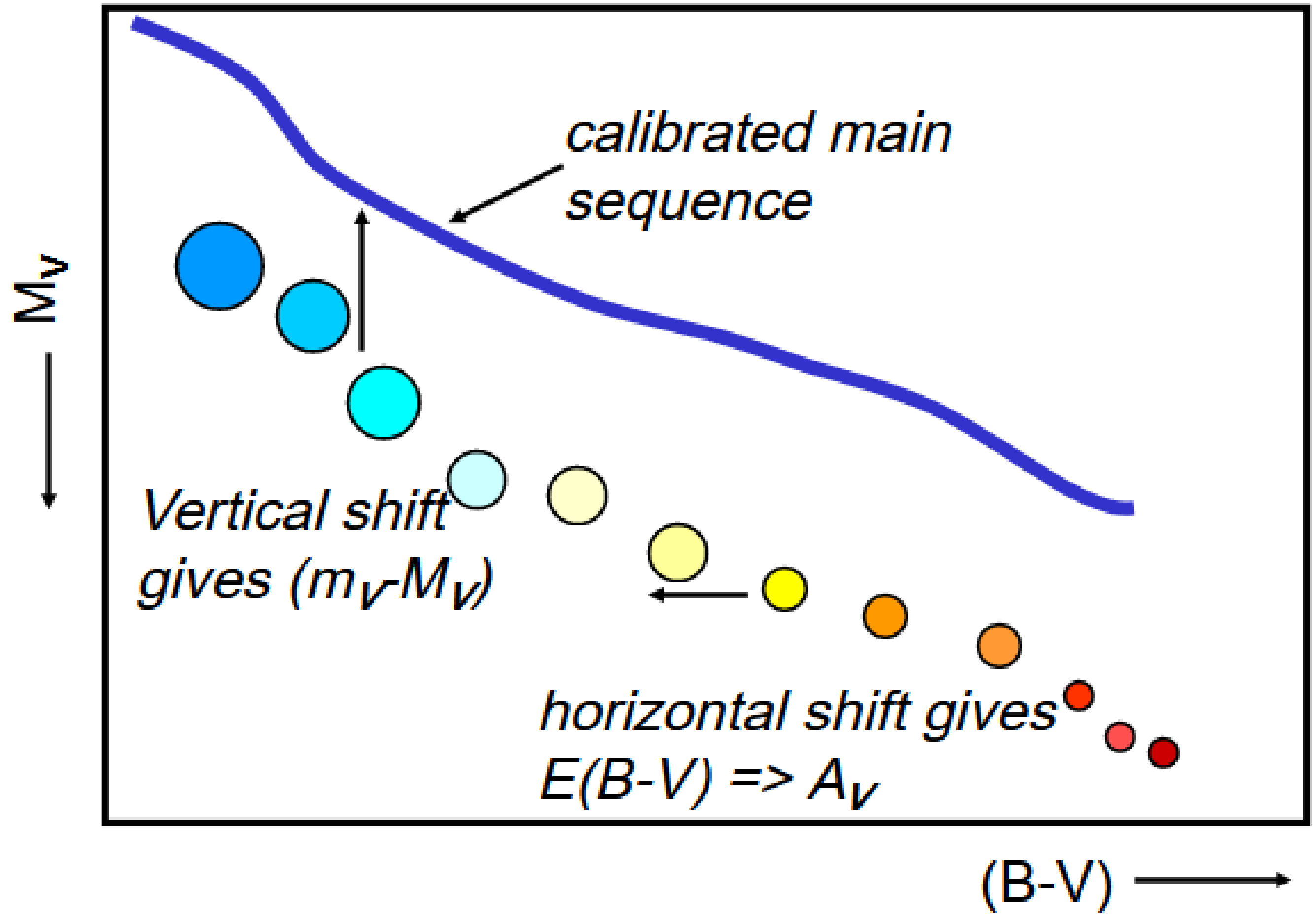




$$b = \frac{L}{(4\pi d^2)}$$

Shift → distance of the clusters.

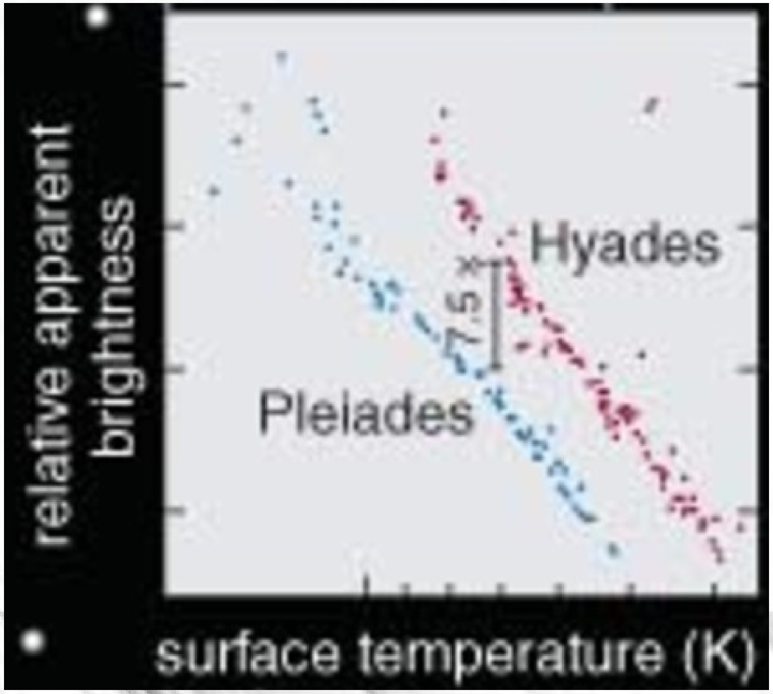
1B11 Cluster distances from H-R diagrams

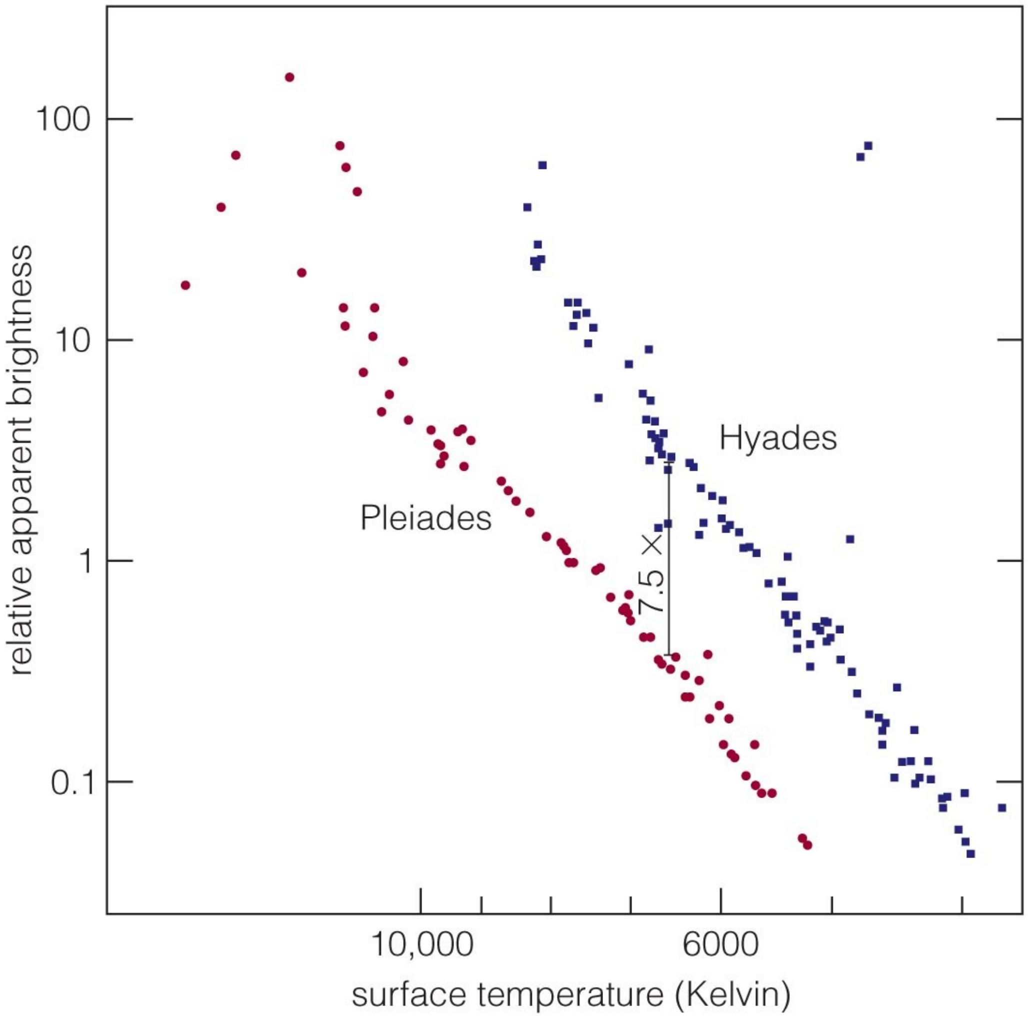


The Hyades are a star cluster located in Taurus constellation and the nearest open cluster of stars to the solar system. lies at a distance of 150 light years, or 47 parsecs, from Earth

the distance to the Pleiades is about 400 light-years from Earth/

The ratio between the distances is 400/150
Or about 2.5. So the one farther away
Should appear 2.5^2 (about 7.5) less bright





Step 3

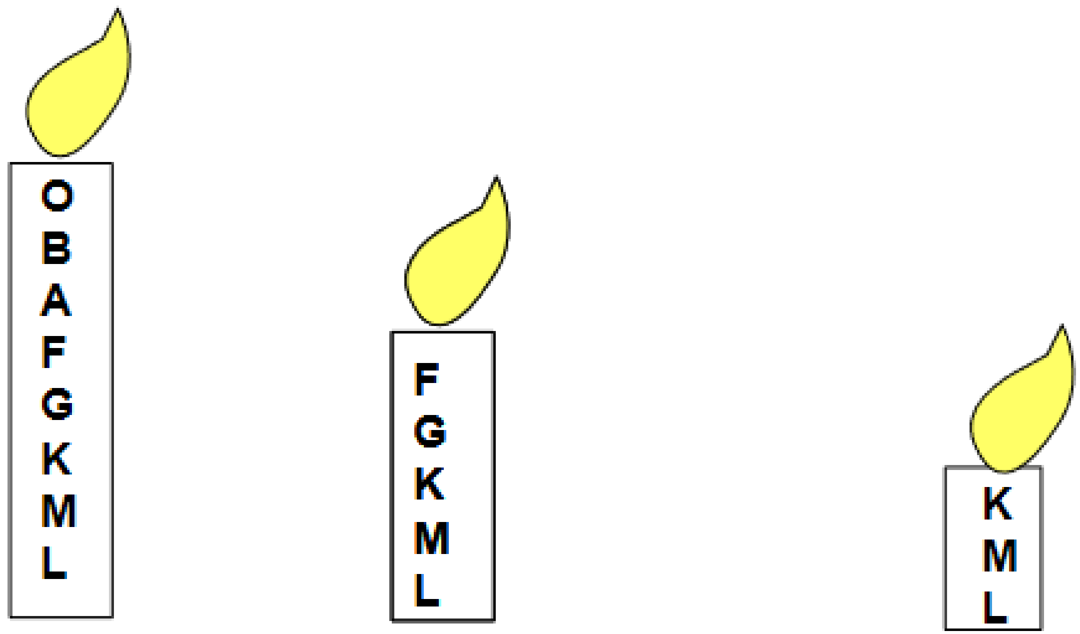
The apparent brightness of a star cluster's main sequence tells us its distance.

So we know the parallax angle of the Hyades = 0.02 arcseconds.
distance(pc) = 1/0.02 = 50 pc . Multiply by 3.26 = 163.07 light years.
So distance Pleiades = 163 x 2.5 = 446 light years away.

7.5 = square root of 2.5 .

We can find the age of the cluster thanks to the HR diagram.
If the cluster is young the hot stars are still here (few million years)
If the cluster is old , hot stars are gone . The turning point → age.

Mass ($M_{\odot}$)	Spectral class	Main-sequence lifetime (10^6 years)
25	O	4
15	B	15
3	A	800
1.5	F	4500
1.0	G	12,000
0.75	K	25,000
0.50	M	700,000

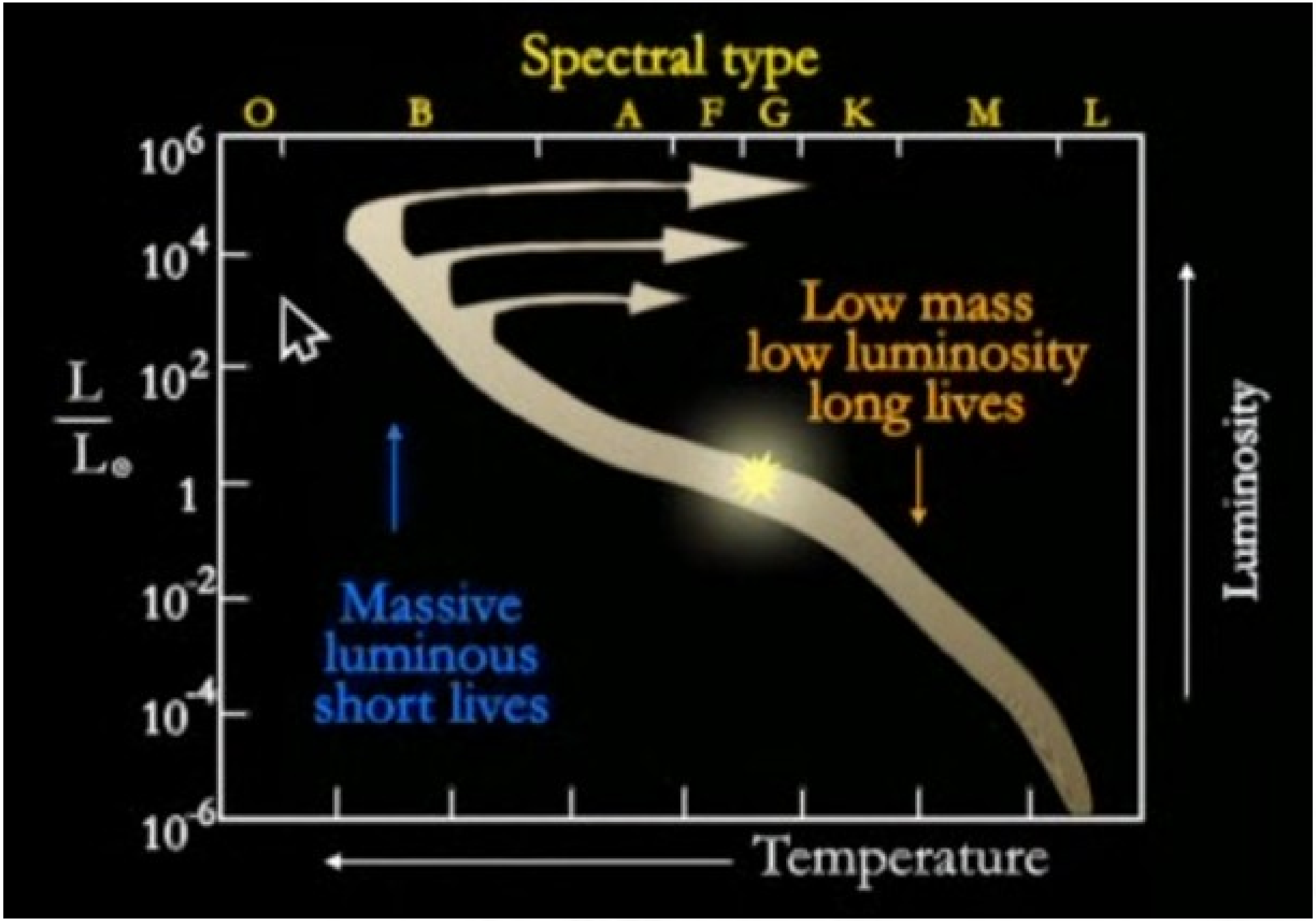


Young cluster

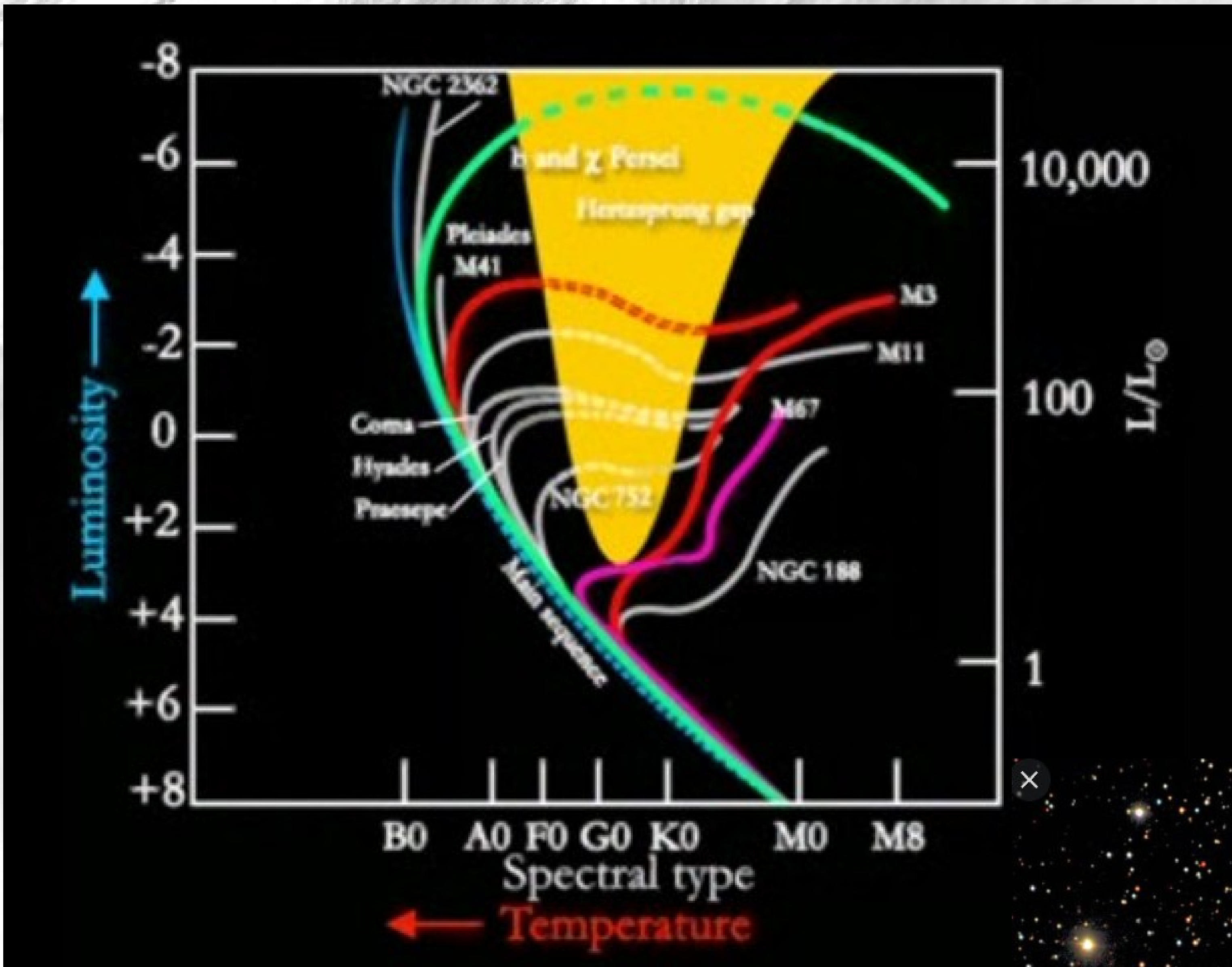
1 billion years old

10 billion years old (no more sun like star)

Look at the most massive star left in
A cluster to find the turning point and
Therefore the age of the cluster.
Further down is the turning point, the older is the cluster.

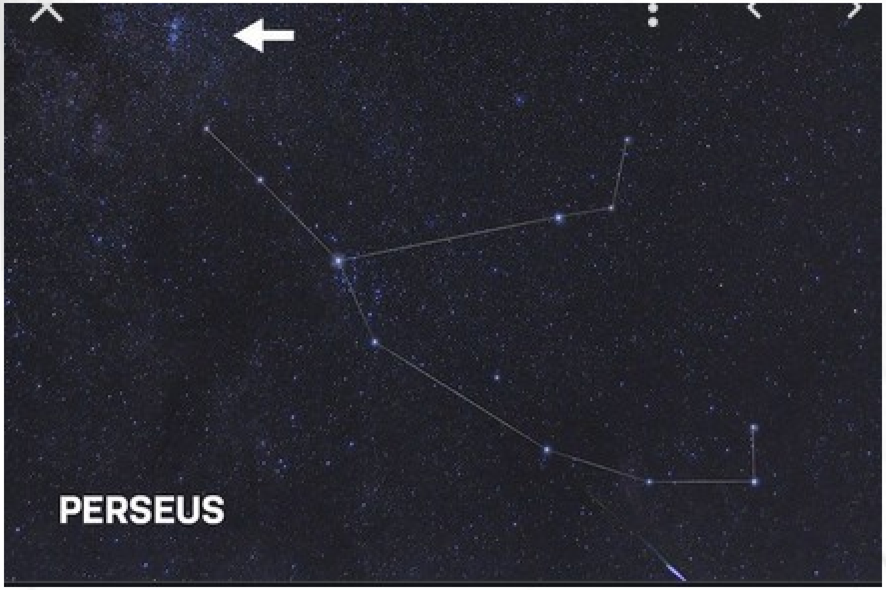


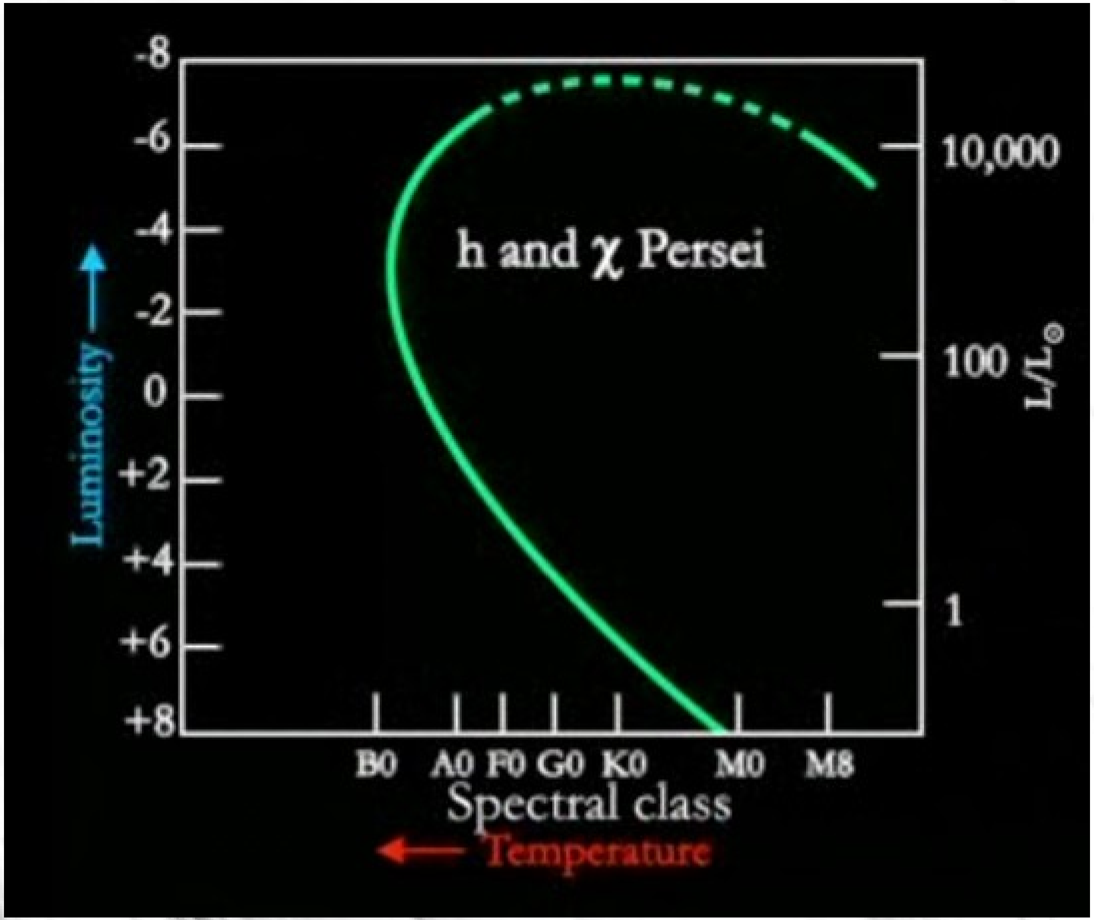
Spectral type
O B A F G K M L
10^6
10^4
10^2
1
10^{-2}
10^{-4}
10^{-6}
$\frac{L}{L_\odot}$
Low mass
low luminosity
long lives
Massive
luminous
short lives
Luminosity
Temperature

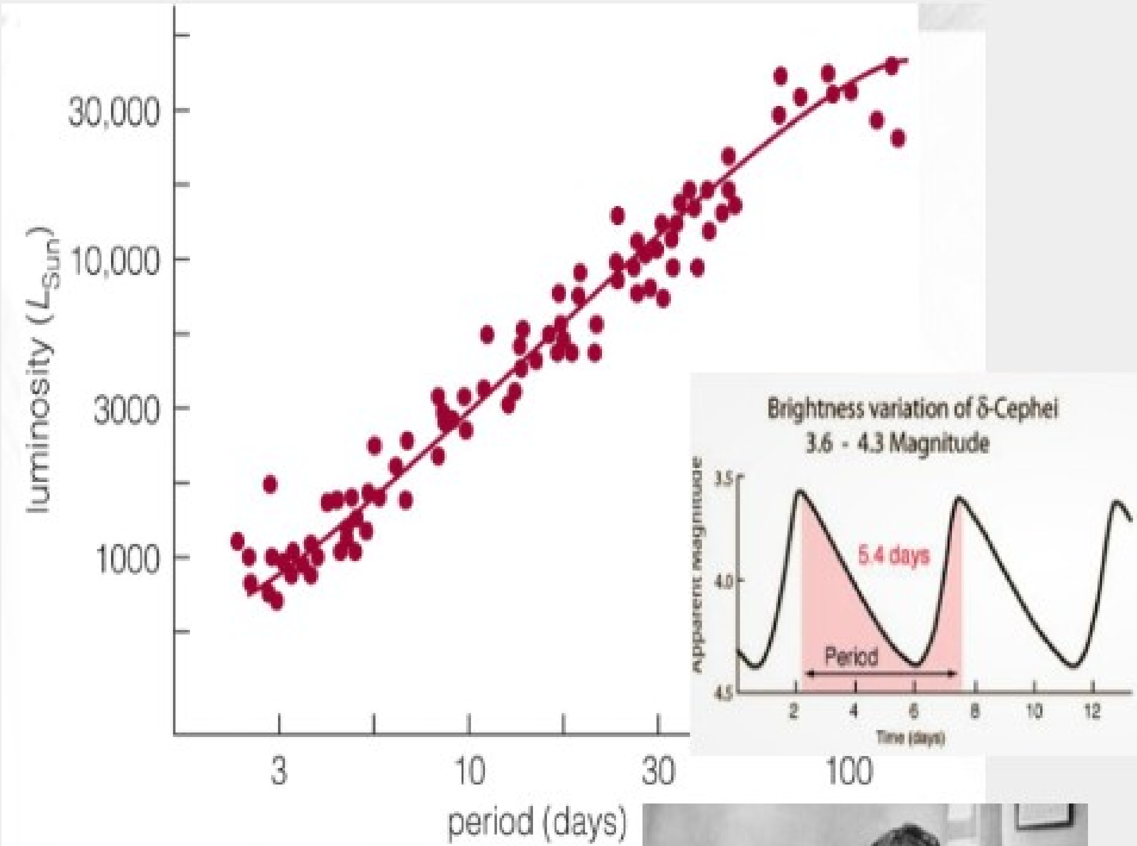


NGC 188 in Cepheus is 6.5 billion years of age

The Double Cluster in Perseus

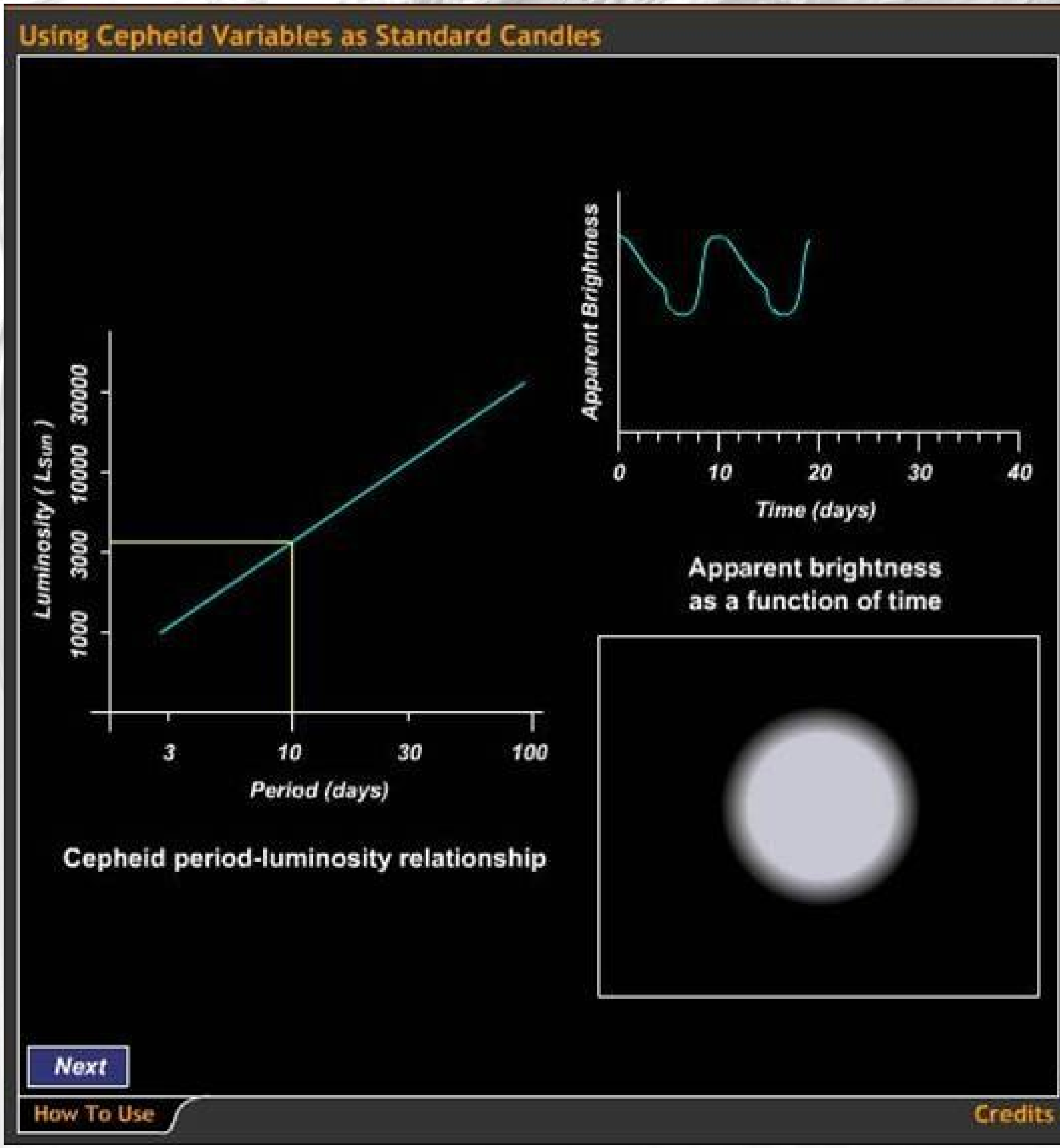






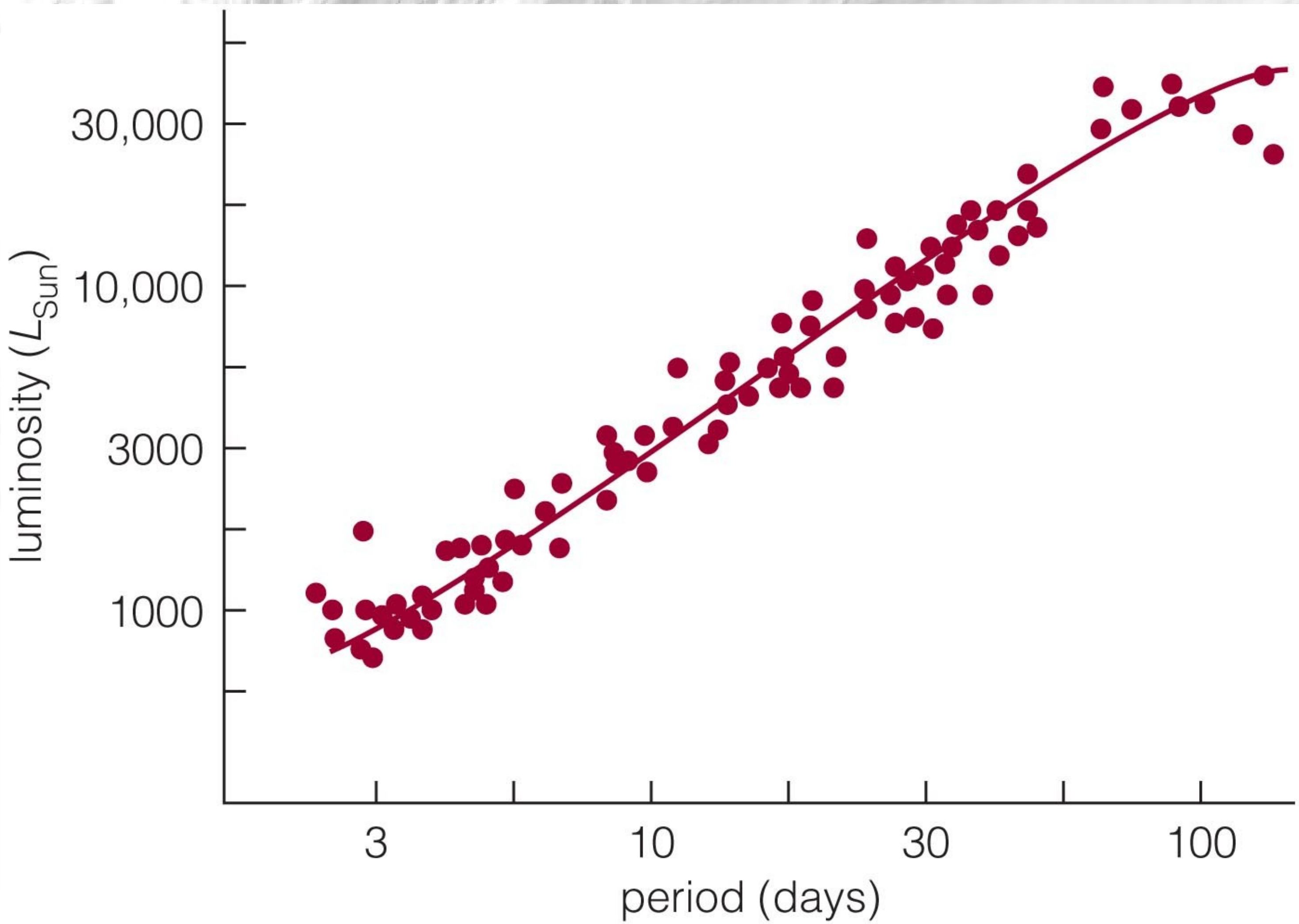
Henrietta Swan Leavitt

Henrietta Swan Leavitt was an American astronomer. A **graduate of Radcliffe College**, she worked at the Harvard College Observatory as a "computer", ...

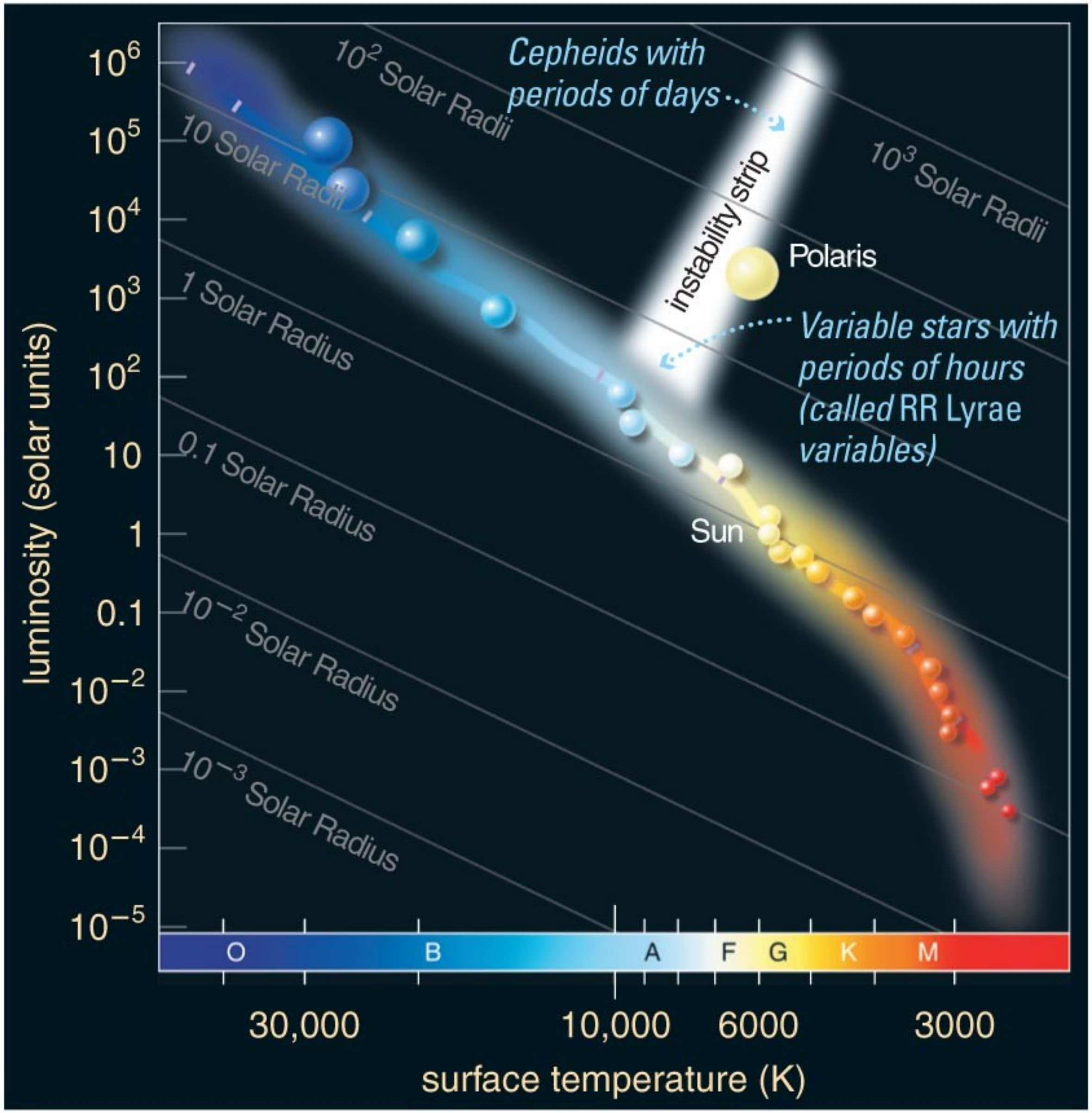


Step 4

Because the period of Cepheid variable stars tells us their luminosity, we can use them as standard candles.



Cepheid variable stars with longer periods have greater luminosities.

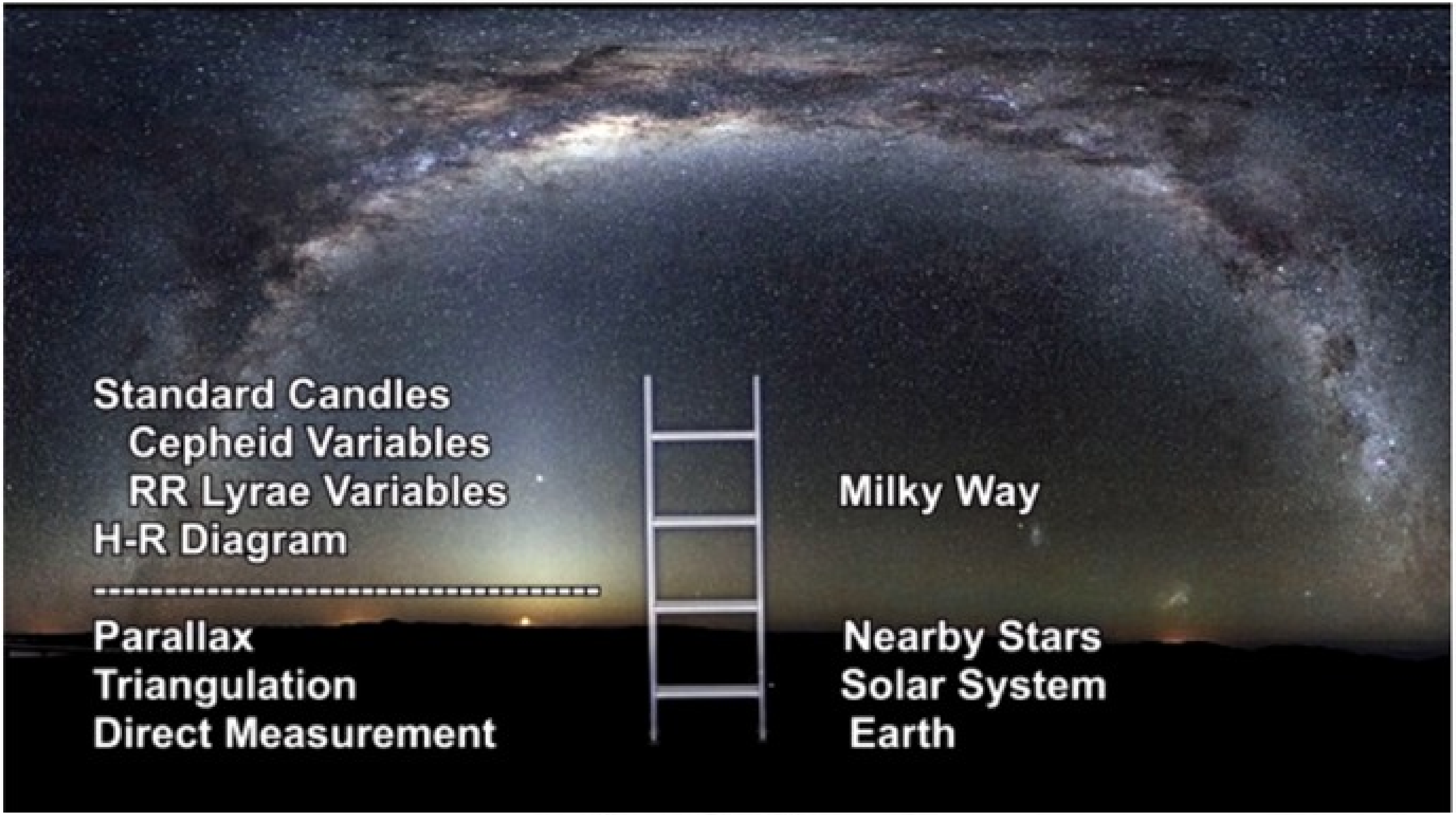


Cepheid variable stars are very luminous.

Henrietta Leavitt

Ejnar Hertzsprung took Leavitt's period-luminosity relationship and applied it to Cepheids in our galaxy where distances could be determined by parallax. This allowed him to convert the relative brightnesses of Cepheids in the Magellanic Clouds to absolute magnitudes, effectively calibrating the relationship.

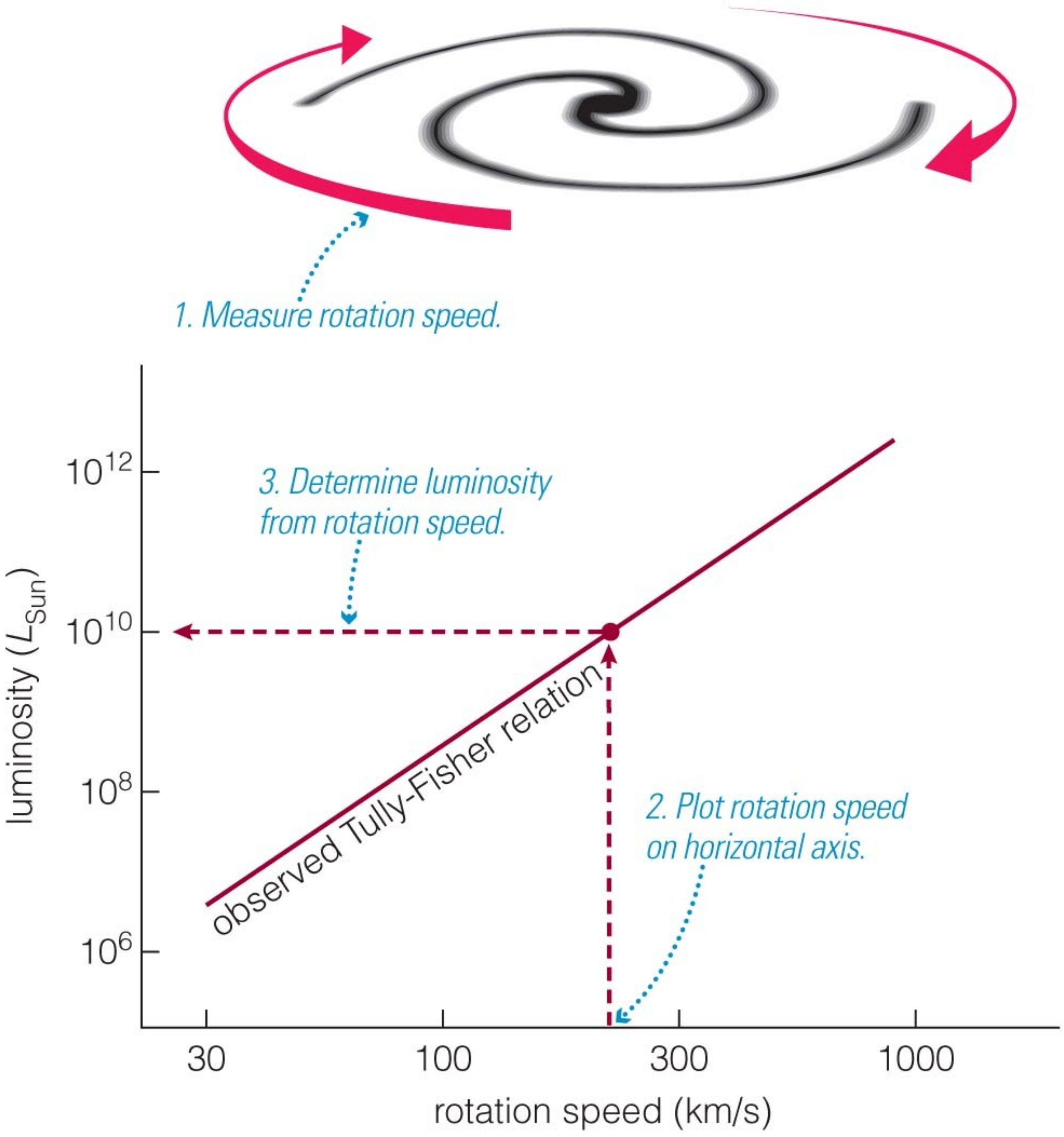
Magellanic clouds studied by Henrietta'


Standard Candles
Cepheid Variables
RR Lyrae Variables
H-R Diagram
Milky Way
Parallax
Triangulation
Direct Measurement
Nearby Stars
Solar System
Earth

Thought Question

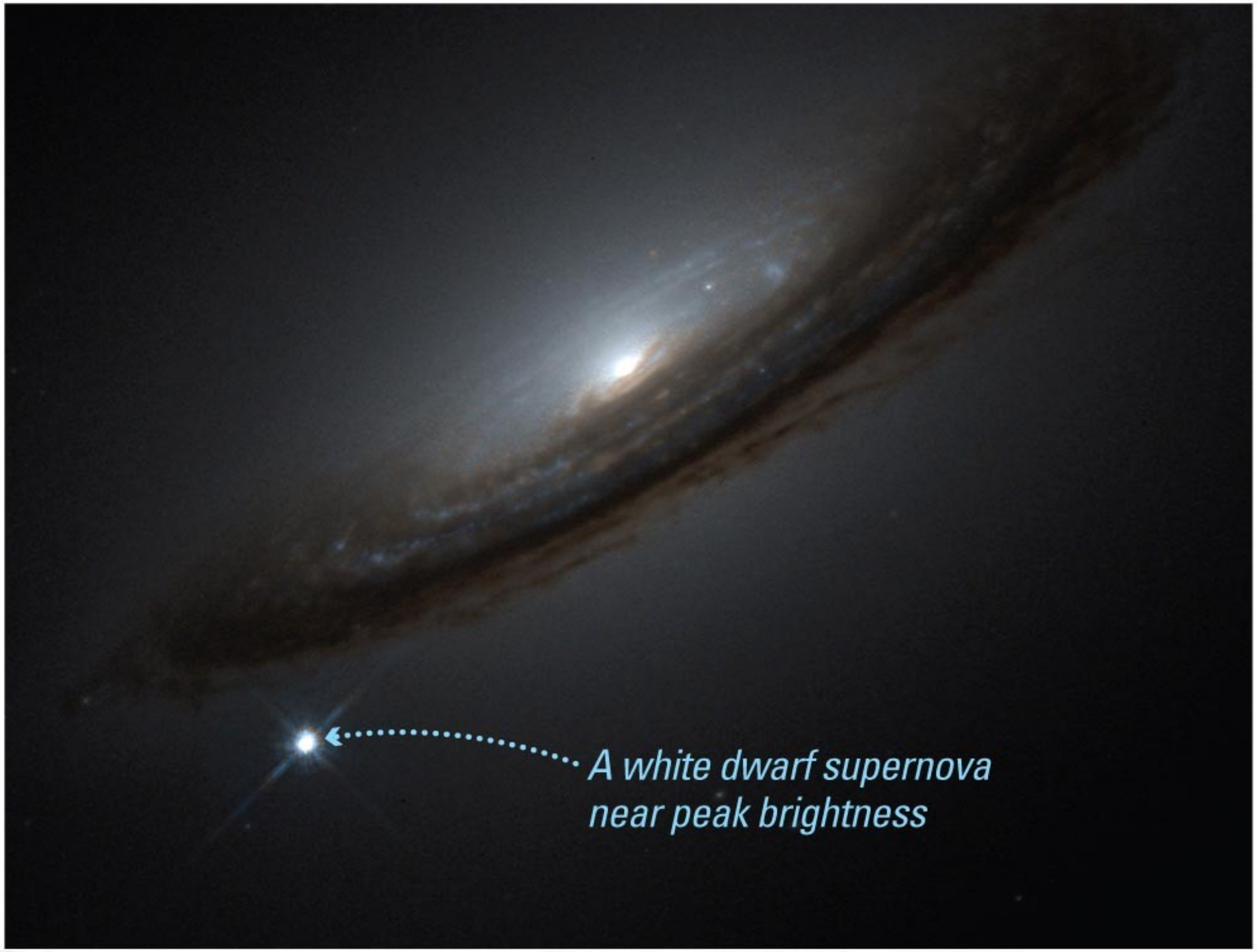
Which kind of stars are best for measuring large distances?

A. high-luminosity stars
B. low-luminosity stars



Tully-Fisher Relation

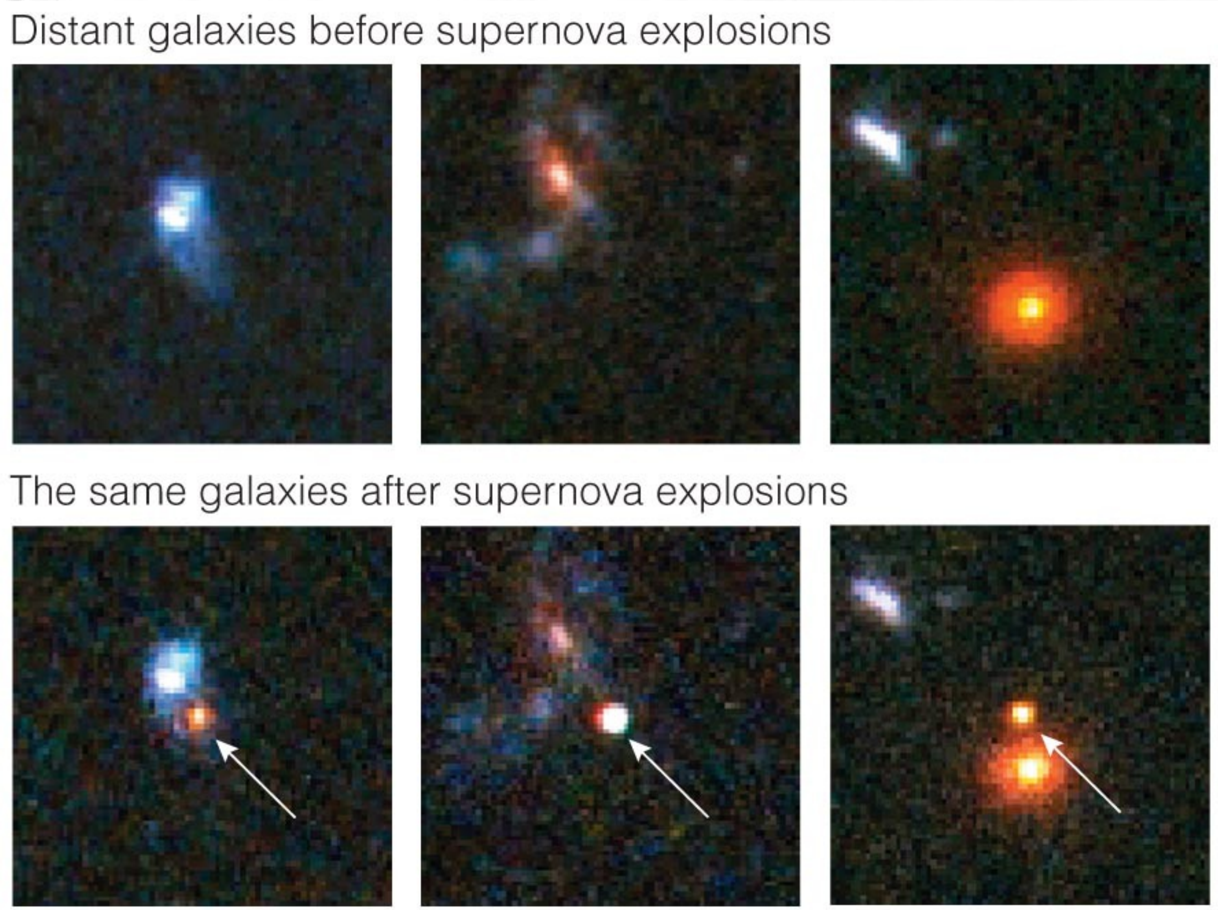
Entire galaxies can also be used as standard candles because a galaxy's luminosity is related to its rotation speed.



White-dwarf supernovae can also be used as standard candles.

For large distances, astronomers use supernovae type IA

Step 5



The apparent brightness of a white dwarf supernova tells us the distance to its galaxy (up to 10 billion light-years).

Weidong Li and Alex Filippenko with the Lick Observatory Katzman Automatic Imaging Telescope (KAIT)

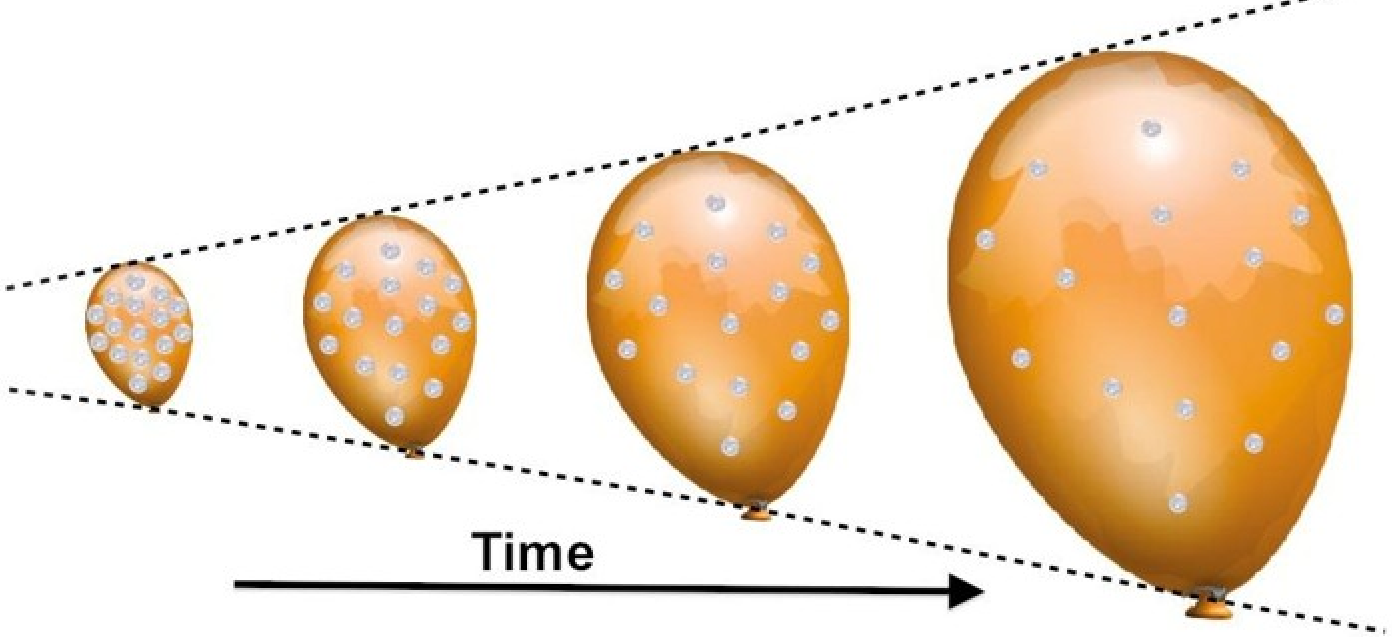
http://articles.adsabs.harvard.edu//full/2001ASPC..246..121F/0000127.000.html

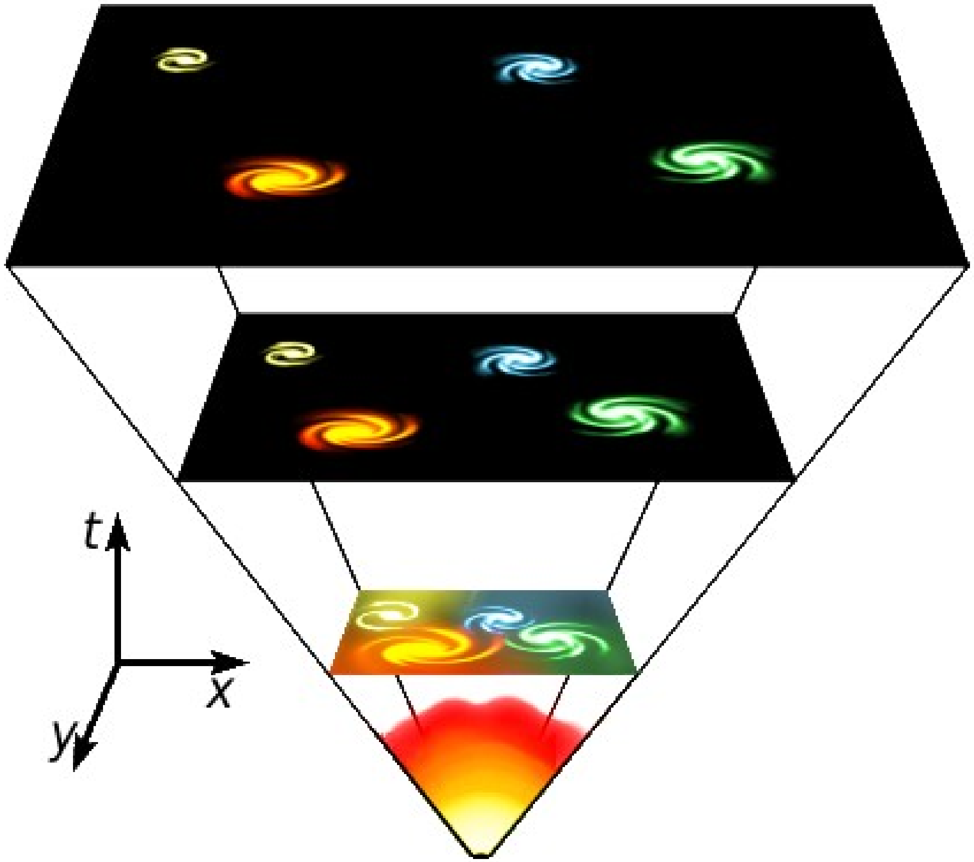
Video
TED_summary.

Hubble law: relationship between distance galaxies and their speed

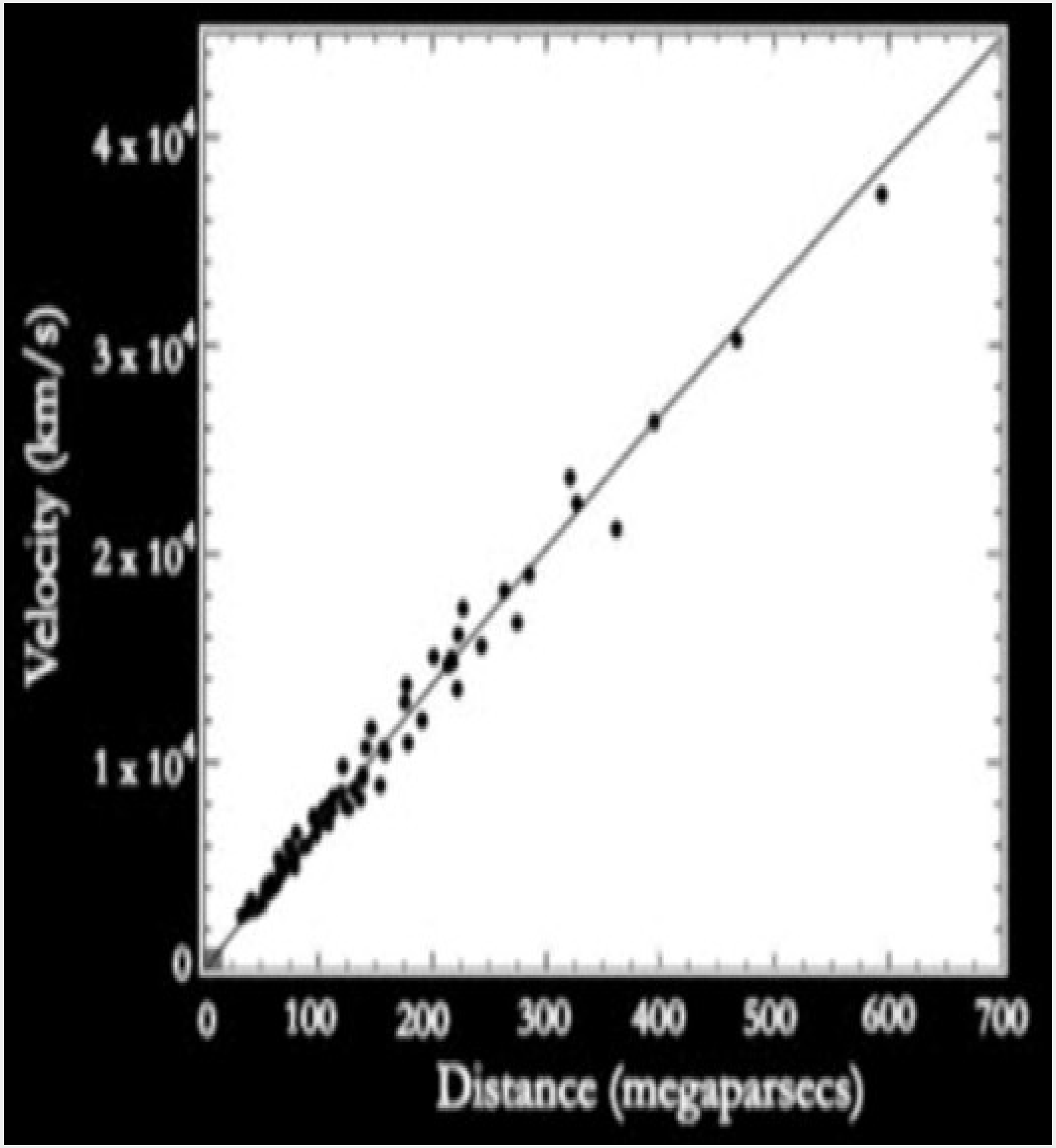
Movie

expansion





Redshift of galaxies → speed
Short movie



2006. Using supernovae as standard candles.

Galaxy Distances from Hubble's Law

$$z = \frac{\Delta\lambda}{\lambda_0} \qquad v \approx zc$$

Hubble's Law: $v = H_0 d$

H_0 = Hubble's constant
c = speed of light
z = redshift
λ = wavelength

H, Hubble's Constant, 70 km/s/Mpc

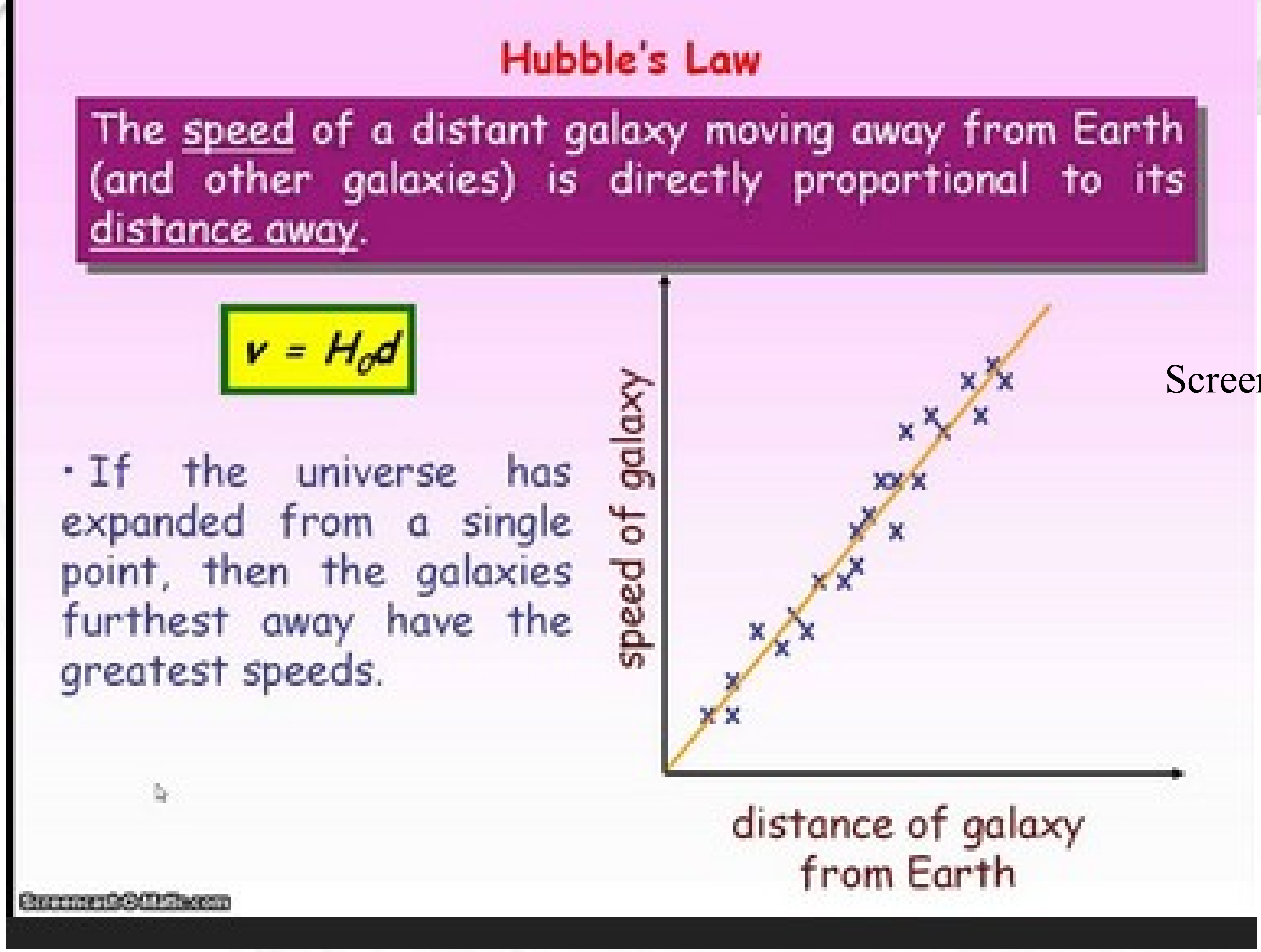
- Mpc = Megaparsec, 3.3 Million Light Years
- Or, H = 21 km/s/MLY   MLY=million light years

d is the distance in Mpc or MLY to the galaxy

Can estimate age of the Universe: $t \sim d/v \sim 1/H \sim 14$ Gyr

$H=2.32\ 10^{-18}\ s^{-1}$

http://hyperphysics.phy-astr.gsu.edu/hbase/Astro/redshf.html#:~:text=Go%20Back-,Measured%20Red%20Shifts,v%2Fc%20%3D



Screenshot Ho

This is the last step of the ladder.

→ With this graph, if we know the relative speed
Of a galaxy → distance.

→ Also. The slope gives us the age of the Universe

Ho is 1/time because V=d/time . time=age Universe

We measure galaxy distances using a chain of interdependent techniques.

The universe is 13.7 billion years old, plus or minus about 130,000 years.